Monopoly Pricing when Consumers Use a Misspecified Model

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Abstract

Consumers often evaluate product quality using a misspecified utility model and learn from market feedback. I study a monopolist's screening problem in which consumers' beliefs are determined by a Berk–Nash fit of a misspecified model under vanishing experimentation, while the monopolist chooses prices under a robust worst-case equilibrium criterion. Misspecification can generate outcomes that do not arise in the correctly specified Mussa–Rosen benchmark, including optimal pooling on the low-quality product. Whether misspecification raises or lowers monopoly profit is governed by two forces: the perceived incentive-compatible markup and the dispersion of perceived utility across equilibrium purchases.

Keywords— Misspecified Models, Berk–Nash Equilibrium, Monopoly Pricing, Second-Degree Price Discrimination, Bounded Rationality

1 Introduction

Although people learn from their past buying experiences, they often find it difficult to predict how much satisfaction a new product will actually bring. Consider a gaming laptop: while 16GB of RAM is clearly more than 8GB, it is much less obvious how much extra satisfaction the additional memory provides. A consumer may therefore rely on a **misspecified model**, such as believing that doubling quality roughly doubles satisfaction, even though the upgrade may make almost no difference to them. Consumers may therefore understand product quality while **misunderstanding how quality translates into utility**.

Learning from experience does not necessarily correct such misspecification. Consumers observe the satisfaction generated by products that are actually purchased, either from their own experience or from the feedback of others. However, they do not observe the relevant counterfactual: how satisfied the same consumer would have been with a different product. As a result, a misspecified model may continue to fit observed experience reasonably well while giving incorrect predictions for unchosen alternatives. What consumers learn therefore depends on what the market allows them to observe.

This dependence of learning on market outcomes changes the standard monopoly pricing problem of Mussa & Rosen (1978). In the classical model, the monopolist sets prices and consumers self-select according to their willingness to pay. Here, prices also affect what consumers learn: they influence which products are purchased, which in turn determines the feedback used to estimate product values. I capture this interaction using **Berk–Nash equilibrium** (Esponda & Pouzo, 2016), where consumers choose optimally given their beliefs while those beliefs are fitted to the outcomes generated by their own choices. The monopolist therefore influences not only demand, but also the **information from which consumers form their willingness to pay**.

On the supply side, the environment **remains close to the standard Mussa–Rosen screening problem**. The monopolist knows the distribution of consumer types, but not the type of any individual consumer, and produces at zero marginal cost. The main complication instead comes from the interaction between pricing and consumer learning. To rule out equilibria sustained only because some products are never tried, I allow a **vanishing amount of experimentation**, so that consumers **receive some feedback about every available product**. If multiple equilibria nevertheless survive, the monopolist evaluates each price menu by its **worst possible equilibrium payoff**. This gives a conservative test of whether the monopolist can profit from consumers' mistaken willingness to pay.

Whether misspecification benefits the monopolist depends on two distinct ways in which it changes the screening problem. The first is the **perceived incentive-compatible markup** between product qualities, which determines how much additional surplus can be extracted from high-type consumers. The second is the **dispersion of perceived utility across consumers' equilibrium purchases**. Misspecification is more favourable to the monopolist when perceived utilities are less dispersed than their true counterparts. Together, these two forces determine whether misspecified learning raises or lowers monopoly profit relative to the correctly specified benchmark.

The role of the incentive-compatible markup becomes particularly important because it depends partly on a **counterfactual valuation**. In a separating outcome, the price premium on the high-quality product is constrained by how much a high-type consumer believes they would gain from choosing it rather than the low-quality alternative. Yet when high-type consumers purchase the high-quality product in equilibrium, their utility from the low-quality product is **never directly observed**. Misspecification can therefore affect profit by distorting not only the value of what consumers buy, but also the value of the alternatives they do not buy.

The dispersion channel instead operates through how consumers interpret the feedback they receive. When perceived utilities across equilibrium purchases are less dispersed than their true counterparts, the misspecified model tends to assign relatively more value to the lower-valued purchase. This gives the monopolist more room to charge a higher price. Thus, a misspecified model closer to **constant utility** tends to create greater scope for profitable exploitation.

A further implication is that the monopolist may benefit from limiting how much consumer experience varies across products. When different types purchase the same product, feedback is generated from a narrower range of product choices, leaving more alternative purchases counterfactual. As a result, consumers have less information about how valuable those alternatives would have been. This provides a new rationale for **pooling**: selling the same product to different consumer types may sacrifice conventional screening opportunities, but can also preserve profitable misspecification.

These mechanisms can move monopoly profit in either direction. Under some misspecified models, distorted beliefs tighten the screening problem and the monopolist earns less than under correct beliefs. Under others, misspecified learning generates sufficiently favourable perceived valuations that the monopolist earns strictly more. Most strikingly, misspecification can make **pooling on the low-quality product** optimal, even though such an allocation is dominated in the correctly specified Mussa–Rosen benchmark. Thus, misspecified learning can change not only the prices charged, but also the **structure of the optimal product menu**.

This paper contributes to the literature on monopoly pricing with boundedly rational consumers by making consumers' perceived willingness to pay **endogenous to market experience**. Rather than imposing a fixed bias in valuation, consumers repeatedly fit a potentially misspecified model to the feedback generated by their equilibrium purchases. This links the Berk–Nash learning framework of Esponda & Pouzo (2016) to the classical screening problem and creates a feedback channel from the monopolist's pricing decision to consumers' subsequent beliefs. The resulting distortion is therefore shaped jointly by the **misspecified model and the market outcomes it generates**.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 presents the model, the Berk–Nash learning process, and the equilibrium refinement. Section 4 develops a series of examples to illustrate when misspecification raises or lowers monopoly profit, before deriving a more general result on the role of utility dispersion. Section 5 discusses directions for further research.

2 Literature Review

This paper is related first to the literature on **behavioural industrial organization**, which studies how firms respond when consumers depart from the standard rational model. Spiegler (2011) studies a range of pricing problems in which consumers have non-standard beliefs or preferences, while Heidhues & Kőszegi (2018) provide a broader survey of how rational firms interact with consumers who make systematic mistakes when evaluating products. A common theme in this literature is that firms may adjust prices, product design, or information provision in response to predictable consumer errors. My paper shares this focus on monopoly pricing against non-standard consumers, but the distortion in willingness to pay is not imposed as a fixed behavioural bias. Instead, consumers learn from market feedback through a potentially misspecified utility model, so their perceived valuations depend on the market outcomes generated in equilibrium.

The paper is also related to **behavioural contract theory**, where a principal designs contracts for agents whose decision making departs from the standard model. Kőszegi (2014) surveys this literature, including screening, mechanism design, and contracts designed to exploit agents' mistakes. For example, Eliaz & Spiegler (2006) study a principal facing agents who differ in their ability to anticipate future changes in their preferences. The principal can screen these differences in sophistication and offer contracts that exploit more naive agents. In my setting, the monopolist similarly designs a menu for consumers whose decisions may be systematically distorted, but consumers differ in their underlying valuation type rather than in their degree of sophistication. The distortion instead comes from the model through which consumers interpret their consumption experience.

A particularly related recent contribution is Clyde (2025), who studies mechanism design when agents make **narrow causal inferences** in a multidimensional decision problem. The principal chooses an incentive scheme while agents evaluate the effect of each action separately and fail to account fully for how actions across different dimensions are related. This creates systematic misperceptions of how actions map into consequences, which the principal takes into account when designing the optimal mechanism. The connection to my paper is that the designer faces agents with a misspecified model of the consequences of their choices. The source of misspecification is different, however: consumers in my model misspecify how product quality and consumer type map into utility, and their beliefs are disciplined by the feedback generated through their equilibrium purchases.

Several papers in behavioural industrial organization study more directly how firms can **exploit mistaken consumers**. Heidhues et al. (2017) consider consumers who neglect additional prices unless these are revealed and show that such mistakes can make inferior products particularly profitable. More recently, Ispano & Schwardmann (2023) study firms' quality disclosure and pricing when consumers are "cursed" in the sense that they are insufficiently sceptical about undisclosed product quality. They show that cursed consumers can be exploited even in the presence of competition, and that policies such as mandatory disclosure, third-party disclosure, or consumer education need not eliminate this exploitation. These papers share with mine the idea that incorrect consumer inference can create profitable opportunities for firms. In contrast, the mistake in my model concerns the utility mapping itself and evolves through learning from realized consumption feedback rather than through a fixed failure to infer hidden prices or undisclosed quality.

Murooka & Yamashita (2023) provide another closely related mechanism-design perspective. They study an adverse-selection environment in which no trade would occur with standard Bayesian-rational buyers, but buyers may instead be behavioural. They show that any incentive-feasible mechanism inducing non-trivial trade gives the buyer a strictly negative ex-ante expected payoff, implying that trade with boundedly rational buyers necessarily exposes some behavioural types to losses. Their result highlights the possibility that a seller can exploit deviations from rational inference in an adverse-selection environment. My question is different: rather than asking whether behavioural trade necessarily harms consumers, I compare the monopolist's profit under misspecified learning with the correctly specified Mussa–Rosen benchmark and characterize when the distortion helps or hurts the monopolist.

Finally, our paper builds on the classical literature on monopoly pricing and quality discrimination. In the benchmark model of Mussa & Rosen (1978), a monopolist designs a menu of qualities and prices to screen consumers with heterogeneous valuations. We introduce misspecified consumers into this environment and study how the monopolist's menu affects not only self-selection, but also the experiences from which consumers learn their perceived utility. The learning component of the model builds directly on the **Berk–Nash equilibrium** of Esponda & Pouzo (2016), which provides a framework for agents who learn while restricting attention to a misspecified set of models. The distinctive feature of the present application is that this learning process is embedded inside a monopoly screening problem. Prices affect consumers' product choices, those choices determine the feedback from which the misspecified model is fitted, and the resulting beliefs in turn determine willingness to pay. The monopolist therefore responds not to a fixed behavioural distortion, but to one that is jointly determined by the misspecified model and the market behaviour it generates.

3 The Model

3.1 Consumers

There is a unit mass of consumers. Each consumer has a type $v \in V = \{1, 2\}$. Type 1 constitutes a fraction $\lambda_1 \in (0, 1)$ of the population, while type 2 constitutes the remaining fraction $\lambda_2 = 1 - \lambda_1$.

Each product is characterized by a non-negative integer $m \in X$, where $X = \mathbb{Z}_{\geq 0}$. The product $m = 0$ represents the outside option of no purchase.

3.2 Utility and Satisfaction

The true utility of a type-v consumer from product $m \in X$ is $u^*(m, v)$, where $u^* : X \times V \to \mathbb{R}_+$.

Let $d \in \mathbb{N}$. Consumers' subjective model is parameterized by a nonempty, convex, and compact set $\Theta \subseteq \mathbb{R}^d$. Each parameter $\theta \in \Theta$ determines a perceived utility function $\hat{u}_\theta : X \times V \to \mathbb{R}_+$. We assume that $\hat{u}_\theta(m, v)$ is continuous in θ for every $(m, v) \in X \times V$.

The outside option is normalized such that $u^*(0, v) = \hat{u}_\theta(0, v) = 0$ for every $v \in V$ and $\theta \in \Theta$. Consumers' model is misspecified if

$$u^* \notin \{\hat{u}_\theta : \theta \in \Theta\}.$$

A consumer first chooses a product m and then observes

$$y = u^*(m, v) + \varepsilon,$$

where $\varepsilon \sim N(0, 1)$ is independently and identically distributed across observations. The variable y is a noisy measurement of true utility and is interpreted as the consumer's observed satisfaction.

Consumers believe that, conditional on some $\theta \in \Theta$, satisfaction is generated according to

$$y = \hat{u}_\theta(m, v) + \varepsilon.$$

Thus, consumers correctly understand the distribution of the noise but may misspecify how product quality and consumer type determine expected satisfaction.

We impose the following restrictions on both the true utility function and every utility function in the consumers' subjective model.

Assumption 1 (Utility restrictions)**.** *Every* $u \in \{u^*\} \cup \{\hat{u}_\theta : \theta \in \Theta\}$ *satisfies:*

1. ***Monotonicity.*** *For every* $m \leq m'$ *and* $v \leq v'$*,* $u(m, v) \leq u(m', v)$ *and* $u(m, v) \leq u(m, v')$*.*
2. ***Increasing differences.*** *For every* $m < m'$*,*

$$u(m', 2) - u(m, 2) \geq u(m', 1) - u(m, 1).$$

For some results, we impose an additional regularity condition on the subjective model.

Definition 1. *The subjective model is* ***regular*** *if its parameter can be written as* $\theta = (k, \gamma)$*, with*

$$\Theta = [\underline{k}, \overline{k}] \times \Gamma,$$

where $0 \leq \underline{k} < \overline{k} < \infty$ *and* Γ *is compact, and there exists a family of utility functions* $\{g_\gamma : \gamma \in \Gamma\}$ *such that*

$$\hat{u}_{(k,\gamma)}(m, v) = k g_\gamma(m, v)$$

for every $(k, \gamma) \in \Theta$ *and* $(m, v) \in X \times V$*.*

A regular subjective model is ***single-template*** *if* $d = 1$*. In this case,* k *is the only parameter and there exists a fixed utility function* g *such that*

$$\hat{u}_k(m, v) = k g(m, v).$$

Definition 2. *A regular subjective model is* ***scale-rich*** *if* $g_\gamma(m, v) > 0$ *and*

$$\frac{u^*(m, v)}{g_\gamma(m, v)} \in [\underline{k}, \overline{k}]$$

for every $\gamma \in \Gamma$*,* $m \in X \setminus \{0\}$*, and* $v \in V$*.*

Scale richness ensures that, at any individual product–type observation cell, the scale parameter can be chosen to match true utility exactly. It does not imply that a single parameter fits all observation cells simultaneously, and therefore does not imply correct specification.

3.3 Learning and Berk–Nash Equilibrium

Let $M \subset X$ be a finite set of products available to consumers, with $0 \in M$. A price vector is denoted by $\mathbf{p} = (p_m)_{m \in M}$, where p_m is the price of product m and $p_0 = 0$.

Under any utility function u, the payoff of a type-v consumer who chooses product m is $u(m, v) - p_m$. Given a belief $\mu \in \Delta(\Theta)$, the consumer's perceived expected payoff is therefore

$$\int_{\Theta} \hat{u}_\theta(m, v)\, d\mu(\theta) - p_m.$$

Consumers learn from a common history of market feedback. Each observation contains the consumer's type, the product chosen, and the resulting satisfaction measure. Feedback generated by both consumer types is pooled, so all consumers hold the same belief μ. They therefore learn not only from observations generated by consumers of their own type, although they observe satisfaction only for products that are actually chosen.

Consider a pure consumption profile $\mathbf{m} = (m_1, m_2) \in M^2$, where m_v is the product chosen by type v. Under the true data-generating process, satisfaction observations from type v follow $N(u^*(m_v, v), 1)$. Under parameter θ, consumers instead believe that these observations follow $N(\hat{u}_\theta(m_v, v), 1)$.

The weighted Kullback–Leibler divergence associated with $\mathbf{m}$ is

$$K(\theta \mid \mathbf{m}) = \sum_{v \in V} \lambda_v D_{\mathrm{KL}}\left(N\left(u^*(m_v, v), 1\right) \,\|\, N\left(\hat{u}_\theta(m_v, v), 1\right)\right).$$

The weight λ_v reflects the frequency with which observations from type v appear in the market data. Since the objective and subjective satisfaction distributions are Gaussian with the same unit variance, the weighted Kullback–Leibler divergence can be written as

$$K(\theta \mid \mathbf{m}) = \frac{1}{2} \sum_{v \in V} \lambda_v \left[u^*(m_v, v) - \hat{u}_\theta(m_v, v)\right]^2.$$

Thus, in the present environment, finding the parameter that best fits the observed satisfaction data is equivalent to minimizing a weighted sum of squared prediction errors.

Let $\Theta(\mathbf{m})$ denote the set of parameters in Θ that minimize $K(\theta \mid \mathbf{m})$. The continuity and compactness assumptions imposed in Section 3.2 ensure that $\Theta(\mathbf{m})$ is nonempty. This minimization has a Bayesian-learning interpretation. When consumers repeatedly update their common belief using Bayes' rule while restricting attention to their potentially misspecified model, their beliefs become concentrated on the parameters that best fit the observed market data.

We first focus on pure-strategy Berk–Nash equilibria.

Definition 3 (Pure-strategy Berk–Nash equilibrium)**.** *Fix a price vector* $\mathbf{p}$*. A pair* $(\mathbf{m}, \mu)$*, where* $\mathbf{m} = (m_1, m_2) \in M^2$ *and* $\mu \in \Delta(\Theta)$*, is a pure-strategy Berk–Nash equilibrium if:*

1. ***Subjective optimality.*** *For every* $v \in V$*,*

$$m_v = \max \arg\max_{m \in M} \left\{ \int_{\Theta} \hat{u}_\theta(m, v)\, d\mu(\theta) - p_m \right\}.$$

   Thus, if several products give the consumer the same perceived net payoff, the consumer chooses the product with the highest quality m*.*

2. ***Best fit.*** *The equilibrium belief assigns probability one to the parameters that best fit the data generated by* $\mathbf{m}$*:*

$$\operatorname{supp}(\mu) \subseteq \Theta(\mathbf{m}).$$

Example 1. *Let* $M = \{0, 1, 2\}$*, with* $\lambda_1 = \lambda_2 = 1/2$*. Suppose that true utility is*

$$u^*(m, v) = \min\{m, v\},$$

while consumers use the misspecified model

$$\hat{u}_\theta(m, v) = \theta m v, \qquad \theta \in \Theta = [0, 1].$$

Fix the price vector $\mathbf{p} = (p_1, p_2) = \left(\frac{1}{2}, 2\right)$*.*

To illustrate how a BNE is verified, rather than to characterize the complete set of equilibria, consider the candidate pooling profile $\mathbf{m} = (1, 1)$*. This means both types of consumers pick the low-quality product. We first calculate the parameter that best fits the data generated by this profile and then check whether the resulting belief induces consumers to choose the same profile.*

Under this profile, the two observed product–type pairs are $(m, v) = (1, 1)$ *and* $(1, 2)$*. True utility equals* 1 *at both pairs, whereas perceived utility equals* θ *and* 2θ*, respectively.*

Since the satisfaction distributions are normal with the same variance, the WKLD minimization is equivalent to

$$\min_{\theta \in [0,1]} \left\{ \frac{1}{4}(1 - \theta)^2 + \frac{1}{4}(1 - 2\theta)^2 \right\}.$$

The unique best-fitting parameter is

$$\theta^* = \frac{3}{5}.$$

The equilibrium belief is therefore degenerate at $\theta^* = 3/5$. *Given this belief and the fixed prices, the perceived net payoffs of the two types are*

	$m = 0$	$m = 1$	$m = 2$
$v = 1$	0	$\frac{3}{5} - \frac{1}{2} = \frac{1}{10}$	$\frac{6}{5} - 2 = -\frac{4}{5}$
$v = 2$	0	$\frac{6}{5} - \frac{1}{2} = \frac{7}{10}$	$\frac{12}{5} - 2 = \frac{2}{5}$

Both types strictly prefer product 1. *The consumption profile generated by the fitted belief is therefore precisely the conjectured profile* $\mathbf{m} = (1, 1)$. *It follows that*

$$(\mathbf{m}, \mu) = \big((1, 1), \delta_{3/5}\big)$$

is a pure-strategy BNE at $\mathbf{p} = (1/2, 2)$.

Other candidate profiles can be checked using the same procedure. For example, under the separating profile $\mathbf{m} = (1, 2)$, *where the low-type consumers pick the low-quality product and the high-type consumers pick the high-quality product, the unique best-fitting parameter is* $\theta^* = \frac{9}{17}$.

At this belief, the perceived net payoff of type 2 *from product* 1 *is*

$$\frac{18}{17} - \frac{1}{2} = \frac{19}{34},$$

whereas its perceived net payoff from product 2 *is*

$$\frac{36}{17} - 2 = \frac{2}{17}.$$

Type 2 *therefore chooses product* 1, *contradicting the candidate profile* $(1, 2)$. *Hence,* $(1, 2)$ *is not a BNE at the given price vector.*

The price vector is treated as fixed in this example, consistently with the definition of BNE. The monopolist's choice of prices is introduced in the next subsection.

3.3.1 Vanishing Experimentation

An ordinary Berk–Nash equilibrium may be sustained by predictions about products that are never chosen and therefore generate no feedback. These predictions may be inherited from consumers' prior beliefs rather than disciplined by market data. We introduce a vanishing amount of exogenous experimentation to address this issue.

Let $\eta = (\eta_1, \eta_2)$ denote the vector of type-specific experimentation rates, where $0 < \eta_v < 1/(|M|-1)$ for every $v \in V$. Given an intended pure choice m_v, the observed choice frequency of type v is

$$\sigma_v^\eta(m \mid m_v) = \begin{cases} 1 - (|M|-1)\eta_v, & \text{if } m = m_v, \\ \eta_v, & \text{if } m \neq m_v. \end{cases}$$

Thus, every product is chosen with positive probability by every type whenever $\eta_v > 0$ for all $v \in V$. Equivalently, a positive fraction of each type experiments with every product, while the remaining consumers make the intended pure choice.

For a pure consumption profile $\mathbf{m} = (m_1, m_2)$, define the perturbed weighted Kullback–Leibler divergence by

$$K^\eta(\theta \mid \mathbf{m}) = \sum_{v \in V} \lambda_v \sum_{\tilde{m} \in M} \sigma_v^\eta(\tilde{m} \mid m_v) D_{\mathrm{KL}}\left(N\left(u^*(\tilde{m}, v), 1\right) \| N\left(\hat{u}_\theta(\tilde{m}, v), 1\right)\right).$$

The corresponding set of best-fitting parameters is

$$\Theta^\eta(\mathbf{m}) = \arg\min_{\theta \in \Theta} K^\eta(\theta \mid \mathbf{m}).$$

Definition 4. *Fix a price vector* $\mathbf{p}$ *and an experimentation vector* $\eta = (\eta_1, \eta_2)$ *with* $\eta_v > 0$ *for every* $v \in V$*. A pair* $(\mathbf{m}, \mu^\eta)$*, where* $\mathbf{m} = (m_1, m_2) \in M^2$ *and* $\mu^\eta \in \Delta(\Theta)$*, is an* η*-BNE if the following conditions hold:*

1. ***Subjective optimality.*** *For every* $v \in V$*,*

$$m_v = \max \arg\max_{m \in M} \left\{ \int_\Theta \hat{u}_\theta(m, v)\, d\mu^\eta(\theta) - p_m \right\}. \tag{1}$$

2. ***Best fit.*** *The belief assigns probability one to the parameters that best fit the data generated under experimentation:*

$$\mathrm{supp}(\mu^\eta) \subseteq \Theta^\eta(\mathbf{m}).$$

A sequence $\{\eta^n\}_{n=1}^\infty$, where $\eta^n = (\eta_1^n, \eta_2^n)$, is a **vanishing experimentation path** if $\eta_v^n \downarrow 0$, for every $v \in V$. For example, experimentation may vanish at the same rate for both types,

$$\eta^n = (\delta_n, \delta_n), \qquad \delta_n \downarrow 0,$$

or at different rates, such as

$$\eta^n = (\delta_n, \delta_n^2).$$

In the latter case, experimentation by type 2 vanishes faster than experimentation by type 1.

Definition 5 (Perfect Berk–Nash equilibrium)**.** *Fix a price vector* $\mathbf{p}$ *and a vanishing experimentation path* $\{\eta^n\}_{n=1}^{\infty}$. *A pure-strategy BNE* $(\mathbf{m}, \mu)$ *is a perfect Berk–Nash equilibrium relative to this path if there exists a sequence of beliefs* $\{\mu^n\}_{n=1}^{\infty}$ *such that, for all sufficiently large* n*,* $(\mathbf{m}, \mu^n)$ *is an* η^n*-BNE and*

$$\mu^n \Rightarrow \mu.$$

The refinement is particularly important for the no-trade profile $\mathbf{m} = (0, 0)$. Since no positive product is purchased, consumers learn nothing about θ from market feedback, and no trade may be sustained entirely by their prior beliefs. Vanishing experimentation ensures that every product is occasionally purchased, allowing consumers to learn from the resulting satisfaction data.

A similar, but less extreme, issue can arise under a pooling profile $\mathbf{m} = (\bar{m}, \bar{m})$. Consumers learn from the satisfaction generated by the product that both types purchase, but not from the products that remain unchosen. If several parameters fit the observed satisfaction equally well while making different predictions about the unchosen products, vanishing experimentation generates the additional data needed to distinguish between them.

Experimentation need not vanish at the same rate across consumer types. For example, one type may experiment at rate δ_n while the other experiments at rate δ_n^2. When the on-path data do not uniquely identify θ, the limiting belief may depend on these relative rates. We therefore define pBNE relative to a specified vanishing experimentation path.

The preceding discussion shows that relative experimentation rates can affect which parameter is selected when the equilibrium observations do not uniquely identify θ. There is, however, an important case in which the relative rates do not matter. If the subjective family contains the true utility function, every positive experimentation vector generates observations from all product–type pairs, and a correctly specified parameter achieves zero prediction error at every pair. The following lemma formalises this observation.

Lemma 1 (Correct specification under full-support experimentation)**.** *Define*

$$\Theta^* = \{\theta \in \Theta : \hat{u}_\theta(m, v) = u^*(m, v) \text{ for every } (m, v) \in M \times V\}.$$

Suppose that $\Theta^* \neq \emptyset$*. Then, for every consumption profile* $\mathbf{m}$ *and every experimentation vector satisfying* $\eta_v > 0$ *for every* $v \in V$*,*

$$\Theta^\eta(\mathbf{m}) = \Theta^*.$$

Consequently, every pBNE belief is supported on parameters that correctly predict utility throughout $M \times V$, independently of the consumption profile and the relative rates at which experimentation vanishes. If $\Theta^* = \{\theta^*\}$, then every pBNE belief is degenerate at θ^*.

Proof. Every parameter in Θ^* predicts all observed utilities correctly and therefore achieves zero loss. Any parameter outside Θ^* makes an incorrect prediction for at least one product–type pair; because full-support experimentation gives that pair positive weight, its loss is strictly positive. Hence, $\Theta^\eta(\mathbf{m}) = \Theta^*$, and the conclusions for pBNE beliefs follow immediately. □

Example 2 (Full-support experimentation selects the truth)**.** *Let $M = \{0, 1, 2\}$ and $V = \{1, 2\}$, and suppose*

$$u^*(m, v) = \begin{cases} 0, & m = 0, \\ m + v, & m > 0, \end{cases}$$

while consumers use the two-parameter model

$$\hat{u}_{k,b}(m, v) = \begin{cases} 0, & m = 0, \\ k(m + bv), & m > 0, \end{cases}$$

where Θ is compact and contains both $(k, b) = (1, 1)$ and $(1/2, 3)$.

Consider the profile $\mathbf{m} = (1, 2)$. Without experimentation, equilibrium feedback is fitted perfectly by every parameter satisfying

$$k(1 + b) = 2.$$

For example, both $(k, b) = (1, 1)$ and $(k, b) = (1/2, 3)$ produce zero on-path prediction error.

Full-support experimentation distinguishes between these parameters. In particular,

$$\hat{u}_{\frac{1}{2},3}(2, 1) = \frac{5}{2} \neq 3 = u^*(2, 1), \qquad \hat{u}_{\frac{1}{2},3}(1, 2) = \frac{7}{2} \neq 3 = u^*(1, 2).$$

By contrast, $(k, b) = (1, 1)$ correctly predicts utility at every product–type pair. Indeed, matching utility at both $(1, 1)$ and $(2, 1)$ requires

$$k(1 + b) = 2 \qquad \text{and} \qquad k(2 + b) = 3,$$

which uniquely gives $k = b = 1$. Hence, whenever $\eta_v > 0$ for every $v \in V$, the perturbed fitting problem uniquely selects $(k, b) = (1, 1)$, even when the weights placed on experimental observations are arbitrarily small.

3.4 The Monopolist's Problem

For the remainder of the analysis, let $M = \{0, 1, 2\}$. Write the price vector as $\mathbf{p} = (p_1, p_2) \in \mathbb{R}^2$, maintaining the normalization $p_0 = 0$. The monopolist knows the market primitives and the consumers' subjective model, but does not observe the type of any individual consumer. Production costs are normalized to zero.

Given a price vector $\mathbf{p}$ and a consumption profile $\mathbf{m} = (m_1, m_2)$, the monopolist's profit is

$$\pi(\mathbf{p}, \mathbf{m}) = \lambda_1 p_{m_1} + \lambda_2 p_{m_2}.$$

For each price vector $\mathbf{p}$, let $M^{\mathrm{pBNE}}(\mathbf{p}) \subseteq M^2$ denote the set of consumption profiles that arise in some pBNE at $\mathbf{p}$. The monopolist evaluates a price vector according to the least profitable pBNE that it may generate. Its robust pricing problem is therefore

$$\varPi^{\mathrm{pBNE}} = \sup_{\mathbf{p} \in \mathbb{R}^2} \inf_{\mathbf{m} \in M^{\mathrm{pBNE}}(\mathbf{p})} \pi(\mathbf{p}, \mathbf{m}).$$

This follows the max–min convention commonly used in mechanism design when a mechanism may induce multiple equilibria. Here, the monopolist chooses prices to maximize profit under the least favourable pBNE consumption profile that those prices may generate. The outer problem is written using a supremum because the desired outcome may be approached by a sequence of prices without being attained at the limiting price, while the infimum captures the monopolist's worst-case evaluation across pBNE profiles.

To evaluate the inner infimum, we must determine which consumption profiles can be supported as pBNE at a given price vector. In particular, consumers must be willing to make the choices specified by a candidate profile under a limiting best-fitting belief. These choices can be characterized using the consumers' individual-rationality and incentive-compatibility constraints. To state these constraints, define perceived expected utility under belief μ by

$$\bar{u}_\mu(m, v) = \int_\varTheta \hat{u}_\theta(m, v)\, d\mu(\theta).$$

For a consumption profile $\mathbf{m} = (m_1, m_2)$ to be subjectively optimal under $(\mathbf{p}, \mu)$, the individual-rationality constraint for each type $v \in V$ is

$$\bar{u}_\mu(m_v, v) - p_{m_v} \geq 0. \tag{IR$_v$}$$

The corresponding incentive-compatibility constraints are

$$\bar{u}_\mu(m_v, v) - p_{m_v} \geq \bar{u}_\mu(m, v) - p_m \qquad \text{for every } m \in M \setminus \{0, m_v\}. \tag{IC$_{v,m}$}$$

In particular, under a separating profile with $m_1 < m_2$, the incentive constraints between the two assigned products imply

$$\bar{u}_\mu(m_2, 1) - \bar{u}_\mu(m_1, 1) \leq p_{m_2} - p_{m_1} \leq \bar{u}_\mu(m_2, 2) - \bar{u}_\mu(m_1, 2).$$

A consumption profile $\mathbf{m}$ belongs to $M^{\mathrm{pBNE}}(\mathbf{p})$ only if there exists a belief μ such that the incentive and individual-rationality constraints hold and μ is a limiting best-fitting belief under vanishing experimentation. Market feedback may fail to identify the parameter of the subjective model **uniquely**. In particular, several parameters may fit the observed satisfaction data equally well while making different predictions about products that are not chosen in equilibrium. Because the incentive constraints depend on these counterfactual predictions, different best-fitting beliefs may support different consumption profiles at the same price vector. This weak-identification problem is one source of pBNE multiplicity and motivates the infimum in the monopolist's robust objective.

3.5 Equilibrium Analysis

Remark 1 (Price ordering)**.** *For the robust pricing problem, it is without loss to restrict attention to price vectors satisfying $p_2 \geq p_1$. If $p_1 > p_2$, product 1 is dominated by product 2, since product 2 provides weakly higher perceived utility at a lower price.*

The preceding remark restricts the ordering of prices. We next derive a separate restriction on equilibrium consumption. Since both types hold the same belief and perceived expected utility has increasing differences, product choice must be weakly increasing in consumer type.

Lemma 2 (Monotonicity of pBNE profiles)**.** *Every pBNE consumption profile $\mathbf{m} = (m_1, m_2)$ is weakly increasing:*

$$m_1 \leq m_2.$$

Proof. Both consumer types hold the same belief μ. Therefore, $\bar{u}_\mu$ inherits increasing differences from the perceived utility functions.

Suppose instead that $m_1 > m_2$. Since type 1 chooses m_1, subjective optimality implies

$$p_{m_1} - p_{m_2} \leq \bar{u}_\mu(m_1, 1) - \bar{u}_\mu(m_2, 1).$$

Since type 2 chooses the lower product m_2, the inequality must be strict; otherwise, the max arg max rule would select m_1. Hence,

$$p_{m_1} - p_{m_2} > \bar{u}_\mu(m_1, 2) - \bar{u}_\mu(m_2, 2).$$

These inequalities contradict increasing differences, which requires

$$\bar{u}_\mu(m_1, 2) - \bar{u}_\mu(m_2, 2) \geq \bar{u}_\mu(m_1, 1) - \bar{u}_\mu(m_2, 1).$$

Therefore, $m_1 \leq m_2$. $\square$

Since $M = \{0, 1, 2\}$, Lemma 1 restricts the possible pBNE consumption profiles to

$$(0,0), \quad (0,1), \quad (0,2), \quad (1,1), \quad (1,2), \quad (2,2).$$

4 The Profit Effects of Misspecification

Section 3 characterized the consumption profiles that may arise in pBNE. We now examine whether misspecification raises or lowers the monopolist's robust profit relative to the correctly specified benchmark.

Let Π^* denote the monopolist's optimal profit under correct specification. We ask whether robust profit under misspecification, Π^{pBNE}, can exceed Π^*.

The next result considers a regular subjective model whose parameter bounds are sufficiently wide. When only one consumer type purchases one product, the market generates only one informative on-path observation cell. A sufficiently large upper bound on k ensures that the consumers' fitting problem can select a scale that matches true utility at this cell exactly.

Lemma 3 (High-type-only profiles)**.** *Suppose the subjective model is scale-rich. For $m \in \{1, 2\}$, a pBNE with consumption profile $(0, m)$ cannot generate profit strictly greater than Π^*.*

Proof. Under $(0, m)$, the only informative on-path observation is the high type's utility from product m. Scale richness allows this observation to be fitted exactly, so every limiting best-fitting belief satisfies

$$\bar{u}_\mu(m, 2) = u^*(m, 2).$$

The high type's participation constraint therefore implies

$$p_m \leq u^*(m, 2),$$

and hence

$$\pi(\mathbf{p}, (0, m)) = \lambda_2 p_m \leq \lambda_2 u^*(m, 2).$$

Under correct specification, the monopolist can obtain at least this profit by selling product m at price $u^*(m, 2)$. Therefore,

$$\pi(\mathbf{p}, (0, m)) \leq \Pi^*.$$

□

Lemma 3 shows that the high-type-only profile $(0, 2)$ cannot generate profit above the correctly specified benchmark. Nevertheless, $(0, 2)$ remains relevant when determining the monopolist's optimal robust profit under misspecification, because it may outperform the other misspecified profiles for some values of λ. By contrast, the other high-type-only profile, $(0, 1)$, need not be considered separately when comparing candidate robust profits: the next lemma shows that it can never outperform $(0,2)$ within the same misspecified environment. It will nevertheless be taken into account whenever it can coexist with an intended profile at the same price vector.

Lemma 4. *Under a scale-rich subjective model, the monopolist can guarantee a profit weakly greater than the maximum profit obtainable from the profile* $(0,1)$.

Proof. By scale richness, the purchased product is fitted exactly under either high-type-only profile. Hence, the maximum profit obtainable from $(0,1)$ is

$$(1-\lambda)u^*(1,2).$$

Now make product 1 unavailable and set

$$p_2 = u^*(2,2).$$

By the preceding no-trade condition, $(0,0)$ cannot coexist at this price. Every pBNE profile therefore has either only the high type or both types purchasing product 2. These profiles generate, respectively,

$$(1-\lambda)u^*(2,2) \qquad \text{and} \qquad u^*(2,2).$$

Since $u^*(2,2) \geq u^*(1,2)$, both outcomes generate at least as much profit as can be obtained from $(0,1)$. $\square$

Combining the preceding profile characterization with Lemma 4, the only profiles that may generate the monopolist's optimal robust profit are

$$(0,2), \qquad (1,1), \qquad (1,2), \qquad (2,2).$$

Among these, Lemma 3 shows that $(0,2)$ cannot generate profit above the correctly specified benchmark. It follows that any increase in profit above Π^* must be generated through one of the profiles

$$(1,1), \qquad (1,2), \qquad (2,2).$$

This does not mean that the remaining profiles can be ignored when evaluating a price vector. The profiles $(0,0)$ and $(0,1)$, although they cannot generate the optimal robust profit themselves, may coexist with an intended profile and reduce the monopolist's worst-case profit. The profile $(0,2)$ must likewise remain in the equilibrium analysis because, although it cannot beat Π^*, it may be optimal within the misspecified environment or coexist with another profile at the same price vector.

4.1 Example 1: Log-Linear Misspecification Model

Consider $M = \{0, 1, 2\}$ and $V = \{1, 2\}$ and let $\lambda := \lambda_1$. Set the true utility to

$$u^*(m, v) = \min\{m, v\},$$

so low-type consumers ($v = 1$) obtain no extra benefit from a higher-quality product, whereas high-type consumers ($v = 2$) value quality linearly. This models the case of casual versus committed gamers choosing laptops.

Correctly specified benchmark. If consumers' perceived utility equals u^*, three equilibrium candidates are possible. For each $\mathbf{m} \in \{(2, 2), (1, 2), (0, 2)\}$ the monopolist's profit-maximising prices and the resulting profits (per consumer) are

(1) $\mathbf{p}_{(2,2)} = (1, 1)$, profit $= 1$;

(2) $\mathbf{p}_{(1,2)} = (1, 2)$, profit $= 2 - \lambda$;

(3) $\mathbf{p}_{(0,2)} = (\infty, 2)$, profit $= 2 - 2\lambda$.

Because $2 - \lambda > \max\{1, 2 - 2\lambda\}$ for every $\lambda \in (0, 1)$, the monopolist induces $\mathbf{m} = (1, 2)$, earning $2 - \lambda$ per consumer.

Misspecified model. Consumers believe that utility belongs to the log-linear family

$$\{\hat{u}_\theta(m, v) = \theta m v : \theta \in \Theta\},$$

where

$$\Theta = [0, \overline{\theta}]$$

and $\overline{\theta} \geq 1$. This is a regular single-template subjective model, with θ as its scale parameter and $g(m, v) = mv$.

After choosing m_v, the consumer observes

$$y_{m_v,v} = u^*(m_v, v) + \varepsilon = \min\{m_v, v\} + \varepsilon.$$

Weighted KL minimisation. Since the subjective model has only one parameter, the limiting wKLD objective is a quadratic in θ. For every nonzero consumption profile $\mathbf{m} = (m_1, m_2) \neq (0, 0)$, the best-fitting parameter is uniquely determined by

$$\theta^*(\mathbf{m}, \lambda) = \arg\min_{\theta \in \Theta} \left\{ \lambda \big(\theta m_1 - \min\{m_1, 1\}\big)^2 + (1 - \lambda)\big(2\theta m_2 - \min\{m_2, 2\}\big)^2 \right\}.$$

First-order condition. The first-order condition gives

$$\theta^*(\mathbf{m},\lambda) = \frac{2m_2^2(1-\lambda)+m_1\lambda}{4m_2^2(1-\lambda)+m_1^2\lambda}. \tag{2}$$

Thus, the limiting belief is degenerate at $\theta^*(\mathbf{m},\lambda)$, and perceived utility is

$$\bar{u}_\mu(m,v) = \theta^*(\mathbf{m},\lambda)mv.$$

In particular, when $m_1 = 0$ and $m_2 > 0$,

$$\theta^*(\mathbf{m},\lambda) = \frac{1}{2}.$$

For the no-trade profile $(0,0)$, the limiting on-path objective is constant in θ, so the parameter is not identified by the expression above.

Solution approach (profile-by-profile). Characterising pBNE outcomes directly is cumbersome because a given price vector $\mathbf{p}$ may support several consumption profiles. I therefore proceed indirectly. For each candidate profile $\mathbf{m}$, I first use (2) to calculate its fitted parameter

$$\theta^*_{\mathbf{m}} := \theta^*(\mathbf{m},\lambda).$$

I then derive the incentive and individual-rationality conditions under which $\mathbf{m}$ can be supported as a pBNE.

For each of the four profiles that may generate an optimal robust profit in the misspecified envrionment, I first calculate the revenue-maximising supporting prices. I then check whether any other pBNE profile coexists at those prices. Importantly, a coexisting profile generally generates different market observations and therefore a different fitted parameter. Thus, when testing whether $\mathbf{m}'$ coexists at prices designed for $\mathbf{m}$, its incentive conditions must be evaluated at $\theta^*_{\mathbf{m}'}$, rather than at $\theta^*_{\mathbf{m}}$.

For later reference, (2) gives

$$\theta^*_{(2,2)} = \theta^*_{(0,2)} = \theta^*_{(0,1)} = \frac{1}{2},$$

whereas

$$\theta^*_{(1,2)} = \frac{8-7\lambda}{16-15\lambda} \qquad \text{and} \qquad \theta^*_{(1,1)} = \frac{2-\lambda}{4-3\lambda}.$$

For every $\lambda \in (0,1)$, these fitted parameters satisfy

$$\frac{1}{2} < \theta^*_{(1,2)} < \theta^*_{(1,1)} < 1.$$

In particular,

$$\theta^*_{(1,1)} - \theta^*_{(1,2)} = \frac{6\lambda(1-\lambda)}{(4-3\lambda)(16-15\lambda)} > 0.$$

(1) $\mathbf{m} = (2,2)$.

The pBNE fit is

$$\theta^*_{(2,2)} = \frac{1}{2}.$$

Since the low type has the tighter constraints, it is sufficient to require

$$\mathrm{IR}_L^{(2,2)}: \qquad p_2 \leq \hat{u}_{\theta^*_{(2,2)}}(2,1) = 2\theta^*_{(2,2)} = 1, \tag{3}$$

$$\mathrm{IC}_L^{(2,2)}: \qquad p_2 - p_1 \leq \hat{u}_{\theta^*_{(2,2)}}(2,1) - \hat{u}_{\theta^*_{(2,2)}}(1,1) = \theta^*_{(2,2)} = \frac{1}{2}. \tag{4}$$

The second inequality is weak because, under the max arg max rule, a low-type consumer who is indifferent between products 1 and 2 chooses product 2.

Revenue is maximised by setting $p_2 = 1$. Since product 1 is not purchased, its price does not directly affect revenue. Any $p_1 \geq 1/2$ supports the profile; below, we select $p_1 = 1/2$ to eliminate coexisting profiles.

(2) $\mathbf{m} = (1,2)$.

The pBNE fit is

$$\theta^*_{(1,2)} = \frac{8-7\lambda}{16-15\lambda}.$$

For the low type to purchase product 1, individual rationality requires

$$\mathrm{IR}_L^{(1,2)}: \qquad p_1 \leq \hat{u}_{\theta^*_{(1,2)}}(1,1) = \theta^*_{(1,2)}. \tag{5}$$

The low type must strictly prefer product 1 to product 2, while the high type may be indifferent because the tie-breaking rule selects product 2. Hence,

$$\mathrm{IC}_L^{(1,2)}: \qquad p_2 - p_1 > \hat{u}_{\theta^*_{(1,2)}}(2,1) - \hat{u}_{\theta^*_{(1,2)}}(1,1) = \theta^*_{(1,2)}, \tag{6}$$

$$\mathrm{IC}_H^{(1,2)}: \qquad p_2 - p_1 \leq \hat{u}_{\theta^*_{(1,2)}}(2,2) - \hat{u}_{\theta^*_{(1,2)}}(1,2) = 2\theta^*_{(1,2)}. \tag{7}$$

The high type's individual-rationality constraint is implied by (5) and (7).

The revenue-maximising supporting prices bind (5) and (7):

$$\mathbf{p}_{(1,2)} = \left(\theta^*_{(1,2)}, 3\theta^*_{(1,2)}\right) = \left(\frac{8-7\lambda}{16-15\lambda}, \frac{24-21\lambda}{16-15\lambda}\right).$$

(3) $\mathbf{m} = (0, 2)$.

The pBNE fit is

$$\theta^*_{(0,2)} = \frac{1}{2}.$$

For the low type to choose the outside option, both positive products must give strictly negative perceived net utility:

$$\mathrm{IC}^{(0,2)}_{L,1}: \qquad p_1 > \hat{u}_{\theta^*_{(0,2)}}(1,1) = \frac{1}{2}, \tag{8}$$

$$\mathrm{IC}^{(0,2)}_{L,2}: \qquad p_2 > \hat{u}_{\theta^*_{(0,2)}}(2,1) = 1. \tag{9}$$

For the high type to purchase product 2, its incentive and individual-rationality constraints are

$$\mathrm{IC}^{(0,2)}_{H}: \qquad p_2 - p_1 \leq \hat{u}_{\theta^*_{(0,2)}}(2,2) - \hat{u}_{\theta^*_{(0,2)}}(1,2) = 1, \tag{10}$$

$$\mathrm{IR}^{(0,2)}_{H}: \qquad p_2 \leq \hat{u}_{\theta^*_{(0,2)}}(2,2) = 2. \tag{11}$$

Revenue is maximised by setting $p_2 = 2$. At this price, (10) requires $p_1 \geq 1$. Hence, any $p_1 \in [1, 2]$ supports the profile. Below, we select

$$\mathbf{p}_{(0,2)} = (2, 2)$$

to eliminate coexisting profiles. Conditional on $(0, 2)$ being selected, the resulting profit is

$$2(1 - \lambda).$$

(4) $\mathbf{m} = (1, 1)$.

The pBNE fit is

$$\theta^*_{(1,1)} = \frac{2 - \lambda}{4 - 3\lambda}.$$

For both types to purchase product 1, the binding restrictions are the low type's individual-rationality constraint and the high type's incentive constraint against product 2:

$$\mathrm{IR}^{(1,1)}_{L}: \qquad p_1 \leq \hat{u}_{\theta^*_{(1,1)}}(1,1) = \theta^*_{(1,1)}, \tag{12}$$

$$\mathrm{IC}^{(1,1)}_{H}: \qquad p_2 - p_1 > \hat{u}_{\theta^*_{(1,1)}}(2,2) - \hat{u}_{\theta^*_{(1,1)}}(1,2) = 2\theta^*_{(1,1)}. \tag{13}$$

The remaining low-type IC constraint and high-type IR constraint are implied by (12) and (13).

Revenue is maximised by setting

$$p_1 = \theta^*_{(1,1)} = \frac{2-\lambda}{4-3\lambda}$$

and choosing any

$$p_2 > 3\theta^*_{(1,1)} = \frac{6-3\lambda}{4-3\lambda}.$$

(5) $\mathbf{m} = (0, 1)$.

The pBNE fit is

$$\theta^*_{(0,1)} = \frac{1}{2}.$$

For the low type to choose the outside option rather than product 1,

$$\mathrm{IC}_L^{(0,1)}: \qquad p_1 > \hat{u}_{\theta^*_{(0,1)}}(1, 1) = \frac{1}{2}. \tag{14}$$

For the high type to purchase product 1, it must prefer product 1 to product 2 and weakly prefer it to the outside option:

$$\mathrm{IC}_H^{(0,1)}: \qquad p_2 - p_1 > \hat{u}_{\theta^*_{(0,1)}}(2, 2) - \hat{u}_{\theta^*_{(0,1)}}(1, 2) = 1, \tag{15}$$

$$\mathrm{IR}_H^{(0,1)}: \qquad p_1 \leq \hat{u}_{\theta^*_{(0,1)}}(1, 2) = 1. \tag{16}$$

These conditions also imply that the low type prefers the outside option to product 2.

Coexisting profiles at $\mathbf{p}_{(2,2)}$. Among the revenue-maximising prices supporting $(2, 2)$, choose

$$\mathbf{p}_{(2,2)} = \left(\frac{1}{2}, 1\right).$$

At these prices, $(2, 2)$ satisfies (3) and (4) with equality.

Consider the other candidate profiles, each evaluated using its own fitted parameter. Under $(1, 2)$, the fit changes to $\theta^*_{(1,2)} > 1/2$, so its low-type IC constraint (6) would require

$$p_2 - p_1 > \theta^*_{(1,2)},$$

which is not the case here as $1 - \frac{1}{2} = \frac{1}{2}$. Under $(0, 2)$, the fit returns to $1/2$, but (8) requires $p_1 > 1/2$, which fails. Under $(1, 1)$, the fit becomes $\theta^*_{(1,1)} > 1/2$, and (13) would require

$$p_2 - p_1 > 2\theta^*_{(1,1)},$$

which also fails as $1 - \frac{1}{2} = \frac{1}{2}$. Finally, $(0, 1)$ is excluded by (14), since $p_1 = 1/2$.

Thus, no other nonzero profile coexists at $\mathbf{p}_{(2,2)}$, and the robust profit generated through $(2, 2)$ is

$$\mathit{\Pi}^{\mathrm{pBNE}}_{(2,2)} = 1.$$

Coexisting profiles at $\mathbf{p}_{(1,2)}$. At the revenue-maximising prices

$$\mathbf{p}_{(1,2)} = \left(\theta^*_{(1,2)}, 3\theta^*_{(1,2)}\right),$$

If the profile were $(2,2)$, the fitted parameter would change to $\theta^*_{(2,2)} = 1/2$. Its low-type IR constraint (3) would then require $p_2 \leq 1$, whereas $p_2 = 3\theta^*_{(1,2)} > 1$. Hence, $(2,2)$ cannot coexist.

If the profile were $(0,2)$, the fitted parameter would again be $1/2$. Its high-type IC constraint (10) would require

$$p_2 - p_1 \leq 1.$$

However,

$$p_2 - p_1 = 2\theta^*_{(1,2)} > 1,$$

so $(0,2)$ cannot coexist.

If the profile were $(1,1)$, the fitted parameter would instead be $\theta^*_{(1,1)}$. By (13), this profile requires

$$p_2 - p_1 > 2\theta^*_{(1,1)}.$$

Since

$$\theta^*_{(1,2)} < \theta^*_{(1,1)},$$

the actual markup $2\theta^*_{(1,2)}$ is too small, so $(1,1)$ cannot coexist.

By contrast, under $(0,1)$ the fitted parameter becomes $\theta^*_{(0,1)} = 1/2$. Since

$$\frac{1}{2} < \theta^*_{(1,2)} < 1 \qquad \text{and} \qquad 2\theta^*_{(1,2)} > 1,$$

conditions (14), (15), and (16) all hold. Therefore, $(0,1)$ coexists with $(1,2)$ at the revenue-maximising prices.

To eliminate $(0,1)$ while retaining the largest permissible markup, lower p_1 to $1/2$ and keep $p_2 - p_1 = 2\theta^*_{(1,2)}$. The resulting prices are

$$\tilde{\mathbf{p}}_{(1,2)} = \left(\frac{1}{2}, \frac{1}{2} + 2\theta^*_{(1,2)}\right).$$

The target profile $(1,2)$ continues to satisfy (5)–(7). Profile $(0,1)$ is now excluded because (14) requires $p_1 > 1/2$. Moreover, $(0,2)$ remains excluded because the markup is strictly greater than 1.

The robust profit generated through $(1,2)$ is therefore

$$\begin{aligned}\Pi^{\mathrm{pBNE}}_{(1,2)} &= \lambda\left(\frac{1}{2}\right) + (1-\lambda)\left(\frac{1}{2} + 2\theta^*_{(1,2)}\right) \\ &= \frac{1}{2} + (1-\lambda)\frac{16-14\lambda}{16-15\lambda}.\end{aligned}$$

Coexisting profiles at $\mathbf{p}_{(0,2)}$. Consider

$$\mathbf{p}_{(0,2)} = (2, 2).$$

Under $(2, 2)$, the low-type individual-rationality constraint (3) requires $p_2 \leq 1$, which fails.

Profiles $(1, 2)$ and $(1, 1)$ cannot coexist because their low-type individual-rationality constraints require, respectively,

$$p_1 \leq \theta^*_{(1,2)} < 1$$

and

$$p_1 \leq \theta^*_{(1,1)} < 1,$$

whereas $p_1 = 2$. Profile $(0, 1)$ is also excluded because (16) requires $p_1 \leq 1$.

Thus, $(0, 2)$ is the unique pBNE profile at $\mathbf{p}_{(0,2)} = (2, 2)$, and its robust profit is

$$\Pi^{\text{pBNE}}_{(0,2)} = 2(1 - \lambda).$$

Coexisting profiles at $\mathbf{p}_{(1,1)}$. Set

$$p_1 = \theta^*_{(1,1)} = \frac{2 - \lambda}{4 - 3\lambda}$$

and choose p_2 sufficiently high that no profile involving product 2 can be supported.

Although the target profile has fitted parameter $\theta^*_{(1,1)}$, the candidate profile $(0, 1)$ would generate $\theta^*_{(0,1)} = 1/2$. Since

$$\frac{1}{2} < \theta^*_{(1,1)} < 1,$$

and $p_2 - p_1$ is sufficiently large, all three conditions (14)–(16) hold. Thus, $(0, 1)$ coexists at the revenue-maximising price for $(1, 1)$.

To eliminate $(0, 1)$, lower p_1 to $1/2$ while keeping p_2 sufficiently high. The profile $(1, 1)$ remains supported because $1/2 < \theta^*_{(1,1)}$, whereas $(0, 1)$ is excluded by (14). The robust profit generated through $(1, 1)$ is therefore

$$\Pi^{\text{pBNE}}_{(1,1)} = \frac{1}{2}.$$

For completeness, the no-trade profile $(0, 0)$ does not coexist at any of the three robust price vectors above. Under no trade, vanishing experimentation selects a fitted parameter between $1/2$ and $3/5$. Hence, the high type's perceived utility from product 1 is at least

1, whereas all three robust price vectors set $p_1 = 1/2$. The high type therefore strictly prefers product 1 to the outside option.

Comparative Profits.

The profile-by-profile analysis gives

$$\begin{aligned} \Pi^* &= 2 - \lambda, \\ \Pi^{\text{pBNE}}_{(2,2)} &= 1, \\ \Pi^{\text{pBNE}}_{(1,2)} &= \frac{1}{2} + (1-\lambda)\frac{16 - 14\lambda}{16 - 15\lambda}, \\ \Pi^{\text{pBNE}}_{(1,1)} &= \frac{1}{2}, \\ \Pi^{\text{pBNE}}_{(0,2)} &= 2(1-\lambda). \end{aligned}$$

For the high-type-only profile $(0, 2)$, the fitted parameter is $\theta^*_{(0,2)} = 1/2$, so consumers correctly perceive the high type's utility from product 2:

$$\hat{u}_{\theta^*_{(0,2)}}(2, 2) = u^*(2, 2) = 2.$$

The resulting profit $2(1-\lambda)$ is attained at $\mathbf{p} = (\infty, 2)^1$. As established in Lemma 3, this profile cannot outperform the correctly specified benchmark. Nevertheless, it cannot be omitted from the comparison because it may maximize robust profit for some values of λ. The profile $(0, 1)$ generates at most $1 - \lambda$ and is therefore dominated by $(0, 2)$.

It follows that the monopolist's optimal robust profit under misspecification is

$$\Pi^{\text{pBNE}} = \max\left\{1,\, 2(1-\lambda),\, \frac{1}{2} + (1-\lambda)\frac{16 - 14\lambda}{16 - 15\lambda}\right\}. \tag{17}$$

Figure 1 compares the robust profit generated by each profile with the correctly specified benchmark.

- **Very thick solid line:** Correctly specified profit $2 - \lambda$, generated by the separating profile $(1, 2)$.
- **Solid thin line:** Robust profit from $(2, 2)$ under misspecification.
- **Dashed thin line:** Robust profit from $(1, 2)$ under misspecification.
- **Dotted thin line:** Robust profit from $(1, 1)$ under misspecification.
- **Dash-dotted thin line:** Robust profit from $(0, 2)$ under misspecification.

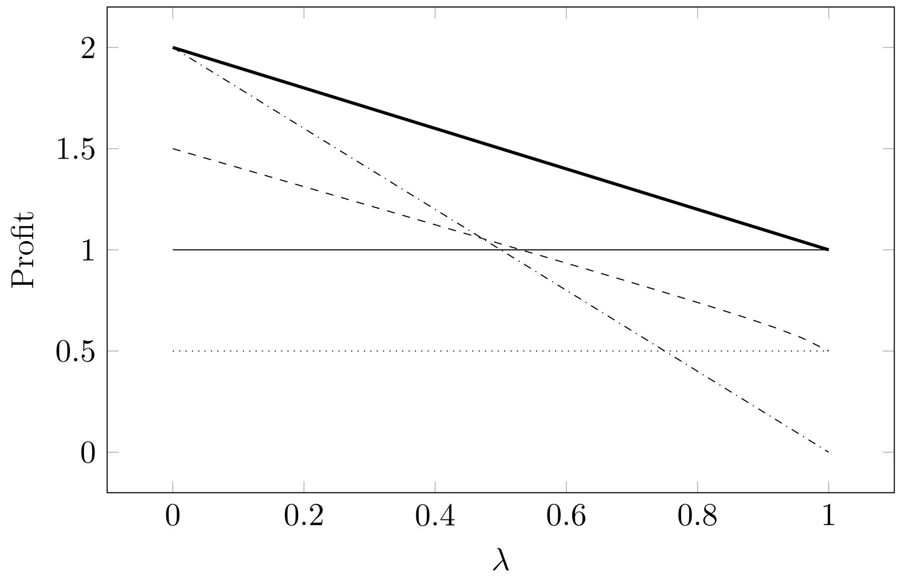

Figure 1: Example 1: Robust profit as a function of λ

Result 1. ***Log-Linear Misspecification Model.*** *For every $\lambda \in (0,1)$, the monopolist's optimal robust profit under log-linear misspecification is strictly lower than its optimal profit under correct specification:*

$$\mathit{\Pi}^{\mathrm{pBNE}} < \mathit{\Pi}^*.$$

Moreover, if

$$\lambda > \frac{45 - \sqrt{233}}{56},$$

the monopolist's optimal robust consumption profile is

$$\mathbf{m} = (2, 2).$$

Thus, when the proportion of low-type consumers is sufficiently large, the monopolist abandons screening and sells the high-quality product to both types. Under correct specification, by contrast, the separating profile $(1, 2)$ is optimal for every $\lambda \in (0, 1)$.

Result 1 establishes two effects of log-linear misspecification. First, the monopolist earns less than under correct specification for every type distribution. Second, misspecification changes the form of the optimal allocation.

[1] p_1 can be set sufficiently high for this profile

Under correct specification, the monopolist can charge the low type 1 for product 1 and the high type 2 for product 2. Separation therefore generates

$$2 - \lambda > 1$$

for every $\lambda \in (0, 1)$, whereas pooling both types on product 2 generates only 1. Hence, pooling is never optimal in the correctly specified benchmark.

Under misspecification, robust separation requires lowering the price of product 1 to $1/2$ in order to eliminate the coexisting profile $(0, 1)$. As the proportion of low types increases, this loss on sales to low-type consumers becomes increasingly costly. By contrast, under $(2, 2)$, the fitted parameter $\theta^*_{(2,2)} = 1/2$ correctly matches both observed utilities, allowing the monopolist to sell product 2 to every consumer at price 1. Once low types are sufficiently numerous, the monopolist therefore prefers to give up screening and pool both types on the high-quality product.

Two Driving Forces Determining Whether a Monopolist Can Gain. The separating profile $(1, 2)$ illustrates two forces governing the profit effect of misspecification.

First, the relative incentive-compatible markup matters. Consider the separating profile $(1, 2)$. The largest possible difference between the two prices, if the monopolist could fully extract the perceived utilities associated with the allocated products, is

$$\hat{u}(2, 2) - \hat{u}(1, 1).$$

Incentive compatibility may prevent the monopolist from capturing this entire difference. The high type's IC constraint instead limits the price markup to

$$p_2 - p_1 \leq \hat{u}(2, 2) - \hat{u}(1, 2).$$

The relevant comparison is therefore not simply the absolute size of the IC markup, but the fraction of the perceived utility difference that can be captured:

$$\frac{\hat{u}(2, 2) - \hat{u}(1, 2)}{\hat{u}(2, 2) - \hat{u}(1, 1)}.$$

Under correct specification, the perceived utilities associated with the allocated products are

$$u^*(1, 1) = 1 \qquad \text{and} \qquad u^*(2, 2) = 2.$$

Moreover,

$$u^*(1, 2) = u^*(1, 1) = 1.$$

Consequently, the monopolist can capture the entire difference between the two perceived utilities:

$$\frac{u^*(2,2) - u^*(1,2)}{u^*(2,2) - u^*(1,1)} = \frac{2-1}{2-1} = 1.$$

Indeed, the correctly specified prices $(p_1, p_2) = (1, 2)$ extract the full utility of both types.

Under log-linear misspecification, the perceived utilities associated with the allocated products are

$$\hat{u}_{\theta^*_{(1,2)}}(1,1) = \theta^*_{(1,2)} \qquad \text{and} \qquad \hat{u}_{\theta^*_{(1,2)}}(2,2) = 4\theta^*_{(1,2)}.$$

The total difference between them is therefore

$$4\theta^*_{(1,2)} - \theta^*_{(1,2)} = 3\theta^*_{(1,2)}.$$

However, the high type perceives product 1 to be worth

$$\hat{u}_{\theta^*_{(1,2)}}(1,2) = 2\theta^*_{(1,2)}.$$

Its IC constraint therefore limits the markup to

$$4\theta^*_{(1,2)} - 2\theta^*_{(1,2)} = 2\theta^*_{(1,2)}.$$

The monopolist can consequently capture only

$$\frac{2\theta^*_{(1,2)}}{3\theta^*_{(1,2)}} = \frac{2}{3}$$

of the perceived utility difference between the allocated products.

Although the absolute IC markup $2\theta^*_{(1,2)}$ is greater than the correctly specified markup 1, this comparison is misleading because the misspecified model also generates a larger perceived utility difference, $3\theta^*_{(1,2)}$. Relative to the perceived surplus available for extraction, the monopolist captures strictly less under misspecification. The remaining $\theta^*_{(1,2)}$ is left to the high type as information rent.

Second, the dispersion in perceived utilities matters. Under the separating profile, low types purchase product 1 and high types purchase product 2. Their true utilities are

$$\big(u^*(1,1), u^*(2,2)\big) = (1,2),$$

whereas the subjective model predicts

$$(\hat{u}_\theta(1,1), \hat{u}_\theta(2,2)) = (\theta, 4\theta).$$

Consumers therefore choose θ by solving

$$\min_{\theta \in \Theta} \left\{ \lambda(1-\theta)^2 + (1-\lambda)(2-4\theta)^2 \right\}.$$

At first glance, it may seem that the fitted value should simply lie somewhere between 1/2, which fits the high type perfectly, and 1, which fits the low type perfectly. However, the quadratic fitting problem places a strong force on θ towards 1/2.

To see why, begin from $\theta = 1$. A small reduction in θ worsens the low-type prediction by the same amount, but moves the high-type prediction 4θ four times as far towards its true value 2. Moreover, because the prediction errors are squared, the sensitivity of the high-type term to θ is governed by $4^2 = 16$, whereas that of the low-type term is governed by $1^2 = 1$. The fitting problem therefore gives much greater effective weight to reducing the error in 4θ.

The first-order condition makes this asymmetry explicit:

$$\lambda(1 - \theta^*_{(1,2)}) = 4(1 - \lambda)(4\theta^*_{(1,2)} - 2).$$

For example, when both types have equal mass, $\lambda = 1/2$, this becomes

$$1 - \theta^*_{(1,2)} = 4(4\theta^*_{(1,2)} - 2),$$

giving

$$\theta^*_{(1,2)} = \frac{9}{17} \approx 0.529.$$

Thus, even when the two types receive equal population weight, the fitted scale lies very close to 1/2. The model almost matches the high type's utility,

$$4\theta^*_{(1,2)} = \frac{36}{17} \approx 2.118,$$

while substantially undervaluing the low type's utility,

$$\theta^*_{(1,2)} = \frac{9}{17} \ll 1.$$

More generally, when the subjective model imposes greater dispersion between the two perceived utilities, the prediction of the larger utility becomes more sensitive to the common scale parameter. Under a quadratic fitting criterion, this gives the fitting problem a stronger incentive to match the larger utility closely, even at the cost of a substantial error in the smaller utility.

This force lowers the perceived utility of product 1 for low types and therefore restricts the price that can be charged to them under separation. At the revenue-maximising prices for $(1, 2)$, the additional coexistence of $(0, 1)$ forces the monopolist to lower this price further to 1/2. Consequently, as low types become more numerous, robust separation becomes less attractive and the monopolist eventually prefers to pool both types on product 2.

Thus, the two forces play distinct roles. The dispersion in perceived utilities determines the fitted scale and the perceived utility difference available to the monopolist. The relative IC markup determines how much of that difference can actually be translated into prices. In this example, the quadratic fit already sacrifices the perceived utility of the low type in order to fit the higher utility more closely, and the IC constraint then prevents the monopolist from capturing the full perceived difference that remains. Together with the presence of coexisting pBNE profiles, this can make screening unprofitable and induce pooling that never occurs in the correctly specified benchmark.

4.2 Example 2: Minimum Misspecification Model

Consider $M = \{0, 1, 2\}$ and $V = \{1, 2\}$, and let $\lambda := \lambda_1$. Set the true utility to

$$u^*(m, v) = mv.$$

Misspecified model. Consumers believe that utility belongs to the minimum family

$$\{\hat{u}_\theta(m, v) = \theta \min\{m, v\} : \theta \in \Theta\},$$

where

$$\Theta = [0, \overline{\theta}]$$

and $\overline{\theta} \geq 2$. This is a regular single-template subjective model, with θ as its scale parameter and

$$g(m, v) = \min\{m, v\}.$$

Low-type consumers therefore believe that they obtain no additional utility from purchasing the high-quality product, whereas their true utility increases linearly with product quality.

This setting can model, for example, a household's purchase of kitchen tools. An average household may believe that only professional chefs obtain substantial additional value from high-end knives and therefore perceive no benefit from purchasing one. In reality, a high-end knife may be considerably more durable and efficient, substantially increasing its utility even for an average household.

Comparative Profits.

Figure 2 compares the profit generated by each profile with the correctly specified benchmark.

- **Very thick solid line:** Correctly specified profit $\max\{2, 4(1 - \lambda)\}$.

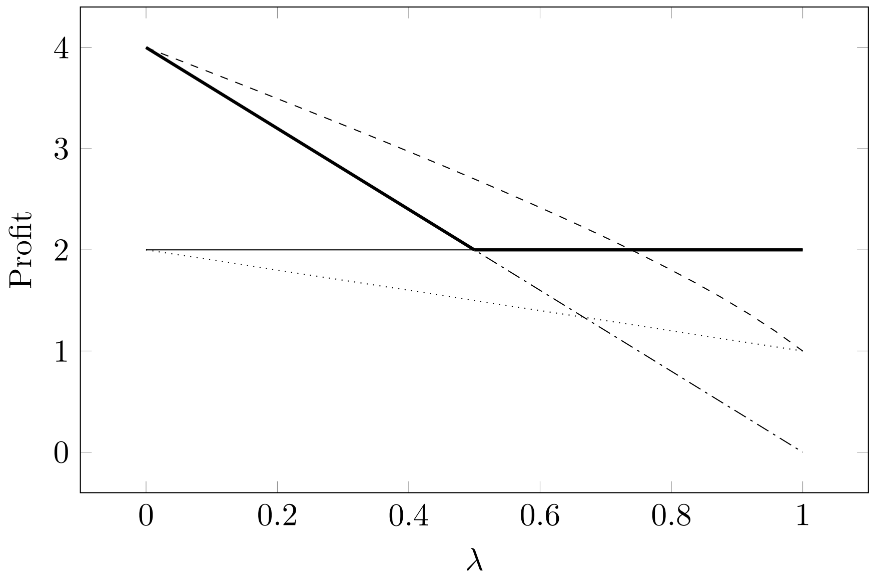

Figure 2: Profit under minimum misspecification and correct specification

- **Solid thin line:** Profit from $\mathbf{m} = (2, 2)$ under misspecification.
- **Dashed thin line:** Profit from $\mathbf{m} = (1, 2)$ under misspecification.
- **Dotted thin line:** Profit from $\mathbf{m} = (1, 1)$ under misspecification.
- **Dash-dotted thin line:** Profit from $\mathbf{m} = (0, 2)$ under misspecification.

Result 2. ***Minimum Misspecification Model.*** *For every*

$$\lambda \in \left(0, \frac{8 - 2\sqrt{2}}{7}\right),$$

the monopolist's optimal robust profit under misspecification is strictly higher than its optimal profit under correct specification.

Unlike the log-linear misspecification model in Section 4.1, the monopolist can exploit consumers' misspecification in this example. Under the separating profile $\mathbf{m} = (1, 2)$, the fitting problem is

$$\min_{\theta \in \Theta} \left\{ \lambda(1 - \theta)^2 + (1 - \lambda)(4 - 2\theta)^2 \right\}.$$

The fitted parameter is

$$\theta^*_{(1,2)} = \frac{8 - 7\lambda}{4 - 3\lambda}.$$

The unexpectedly high utility experienced by high types from product 2 pushes up the fitted scale. This higher scale also raises the perceived utility of product 1 for low types, allowing the monopolist to charge them a higher price.

The profitability of this misspecification can be understood through the same two forces identified in Example 1.

First, the relative incentive-compatible markup matters. Under correct specification, the perceived utilities associated with the two allocated products are

$$u^*(1,1) = 1 \qquad \text{and} \qquad u^*(2,2) = 4.$$

The total difference between them is therefore

$$u^*(2,2) - u^*(1,1) = 3.$$

However, the high type obtains utility

$$u^*(1,2) = 2$$

from product 1. Its incentive constraint therefore limits the markup to

$$u^*(2,2) - u^*(1,2) = 4 - 2 = 2.$$

Thus, under correct specification, the monopolist can capture only

$$\frac{u^*(2,2) - u^*(1,2)}{u^*(2,2) - u^*(1,1)} = \frac{2}{3}$$

of the perceived utility difference between the allocated products.

Under the minimum misspecification model, the corresponding perceived utilities are

$$\hat{u}_{\theta^*_{(1,2)}}(1,1) = \theta^*_{(1,2)} \qquad \text{and} \qquad \hat{u}_{\theta^*_{(1,2)}}(2,2) = 2\theta^*_{(1,2)}.$$

Moreover,

$$\hat{u}_{\theta^*_{(1,2)}}(1,2) = \hat{u}_{\theta^*_{(1,2)}}(1,1) = \theta^*_{(1,2)}.$$

The high type therefore perceives no additional utility from product 1 relative to the low type. Its IC constraint permits the markup

$$2\theta^*_{(1,2)} - \theta^*_{(1,2)} = \theta^*_{(1,2)},$$

which is exactly the full difference between the perceived utilities associated with the allocated products. Hence, the monopolist captures the fraction

$$\frac{\hat{u}_{\theta^*_{(1,2)}}(2,2) - \hat{u}_{\theta^*_{(1,2)}}(1,2)}{\hat{u}_{\theta^*_{(1,2)}}(2,2) - \hat{u}_{\theta^*_{(1,2)}}(1,1)} = 1.$$

Unlike under correct specification, no part of this perceived utility difference must be left to the high type as information rent.

Second, the dispersion in perceived utilities matters. Under the separating profile, the true utilities generated by the two purchases are

$$\big(u^*(1,1), u^*(2,2)\big) = (1,4),$$

whereas the subjective model predicts

$$(\hat{u}_\theta(1,1), \hat{u}_\theta(2,2)) = (\theta, 2\theta).$$

The subjective model therefore imposes less dispersion:

$$\frac{\hat{u}_\theta(2,2)}{\hat{u}_\theta(1,1)} = 2 < 4 = \frac{u^*(2,2)}{u^*(1,1)}.$$

At first glance, the fitted parameter may appear to be an unspecified compromise between $\theta = 1$, which fits the low type perfectly, and $\theta = 2$, which fits the high type perfectly. The quadratic fitting problem, however, creates a strong force towards $\theta = 2$.

Starting from $\theta = 1$, a small increase in θ worsens the low-type prediction by that amount but moves the high-type prediction 2θ twice as far towards its true value 4. Because the errors are squared, the sensitivity of the high-type term to θ is governed by $2^2 = 4$, whereas that of the low-type term is governed by $1^2 = 1$. The fitting problem therefore gives greater effective weight to reducing the error in the larger utility.

The first-order condition is

$$\lambda(\theta^*_{(1,2)} - 1) = 2(1-\lambda)(4 - 2\theta^*_{(1,2)}).$$

For example, when $\lambda = 1/2$,

$$\theta^*_{(1,2)} = \frac{9}{5} = 1.8.$$

Thus, even when the two types have equal mass, the fitted scale lies much closer to 2 than to 1. The model nearly matches the high type's true utility,

$$2\theta^*_{(1,2)} = \frac{18}{5} = 3.6,$$

while substantially overvaluing product 1 for the low type:

$$\theta^*_{(1,2)} = \frac{9}{5} > 1.$$

The smaller dispersion imposed by the subjective model therefore causes the quadratic fit to raise the common scale in order to match the larger utility more closely. This increase spills over to the smaller utility, allowing the monopolist to charge the low type more than its true utility.

Together, the two forces work in the monopolist's favour. The quadratic fit raises the perceived utilities associated with the allocated products, while the subjective model allows the monopolist to capture the entire perceived difference between them through the IC markup. As a result, misspecification can generate prices above the correctly specified screening prices for both types and raise the monopolist's profit above the correctly specified benchmark.

4.3 Example 3: Exponent Misspecification

Consider $M = \{0, 1, 2\}$ and $V = \{1, 2\}$, and let $\lambda := \lambda_1$.

True utility. True utility is

$$u^*(m, v) = \begin{cases} 0, & m = 0, \\ v, & m > 0. \end{cases}$$

Thus, once a consumer purchases a positive product, utility depends only on the consumer's type and not on product quality.

Misspecified model. Consumers believe that utility belongs to the exponent family

$$\{\hat{u}_\theta(m, v) = \theta m^v : \theta \in \Theta\},$$

where

$$\Theta = [0, \overline{\theta}]$$

and $\overline{\theta} \geq 2$. This is a regular single-template subjective model, with θ as its scale parameter and

$$g(m, v) = m^v.$$

Consumers therefore believe that utility is a power function of product quality, with high types perceiving a particularly strong return to quality. In reality, however, quality provides no additional utility once a positive product is purchased.

Consider professional tennis players choosing rackets. They may over-attribute their performance to racket quality, even though their skill determines most of the outcome. Positive feedback from high-type buyers may raise the fitted scale and thereby increase every consumer's perceived value of a low-quality racket. Conversely, when high types purchase premium rackets, their feedback may reveal the limited additional benefit of quality and lower the fitted scale.

Comparative profits.

Figure 3 plots each profit curve as a function of λ.

- **Very thick solid line:** Correctly specified profit $\max\{1, 2(1-\lambda)\}$.
- **Solid thin line:** Robust profit from $\mathbf{m} = (2, 2)$.
- **Dashed thin line:** Robust profit from $\mathbf{m} = (1, 2)$.
- **Dotted thin line:** Robust profit from $\mathbf{m} = (1, 1)$.
- **Dash-dotted thin line:** Robust profit from $\mathbf{m} = (0, 2)$.

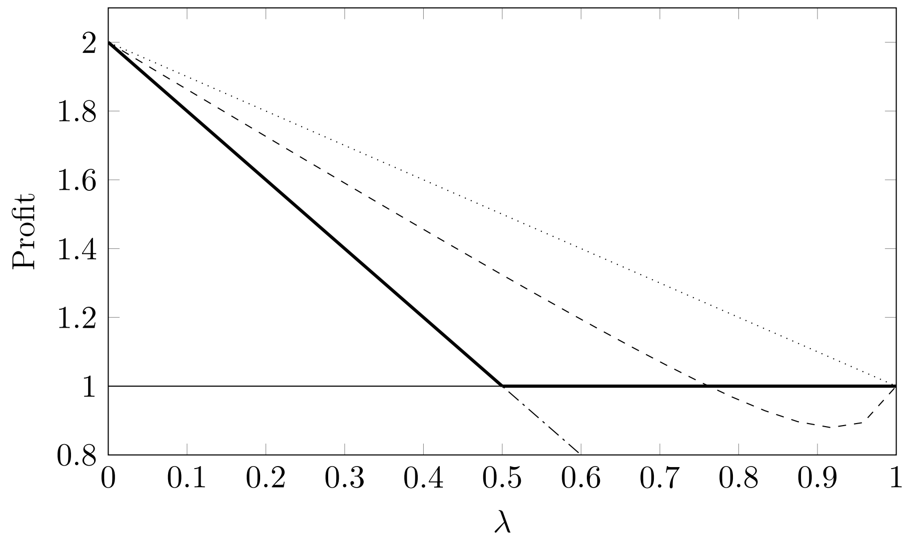

Figure 3: Profit under exponent misspecification and correct specification

Result 3 (Exponent Misspecification)**.** *For every* $\lambda \in (0, 1)$*:*

(i) the monopolist's optimal robust profit under misspecification strictly exceeds its optimal profit under correct specification; and

(ii) the optimal consumption profile under misspecification is $\mathbf{m} = (1, 1)$*, which is never optimal under correct specification.*

Result 3 follows directly from the profit comparisons. The pooling profile $(1,1)$ generates

$$\mathit{\Pi}^{\mathrm{pBNE}}_{(1,1)} = 2 - \lambda.$$

If $\lambda < 1/2$, the correctly specified benchmark is $2(1-\lambda)$, and

$$(2-\lambda) - 2(1-\lambda) = \lambda > 0.$$

If $\lambda \geq 1/2$, the correctly specified benchmark is 1, and

$$(2-\lambda) - 1 = 1 - \lambda > 0.$$

Moreover,

$$(2-\lambda) - \frac{(8-7\lambda)(4-3\lambda)}{16-15\lambda} = \frac{6\lambda(1-\lambda)}{16-15\lambda} > 0.$$

Hence, $(1,1)$ is optimal under misspecification for every $\lambda \in (0,1)$.

Misspecification is profitable because feedback from high types raises the fitted scale under $(1,1)$. This increase spills over entirely to low types, who consequently overvalue product 1 and are willing to pay more than their true utility. The result can again be understood through the two driving forces.

First, the relative incentive-compatible markup matters. Consider first the possibility of screening consumers through the profile $(1,2)$.

Under correct specification, the utilities associated with the two allocated products are

$$u^*(1,1) = 1 \qquad \text{and} \qquad u^*(2,2) = 2.$$

The difference between them is 1. However, the high type obtains the same true utility from either positive product:

$$u^*(1,2) = u^*(2,2) = 2.$$

The high type's IC constraint therefore permits no positive markup for the high-quality product:

$$u^*(2,2) - u^*(1,2) = 0.$$

Thus, under correct specification, the monopolist can capture none of the utility difference between the allocated products through the IC markup:

$$\frac{u^*(2,2) - u^*(1,2)}{u^*(2,2) - u^*(1,1)} = 0.$$

Under exponent misspecification, the perceived utilities associated with the allocated products are

$$\hat{u}_{\theta^*_{(1,2)}}(1,1) = \theta^*_{(1,2)} \qquad \text{and} \qquad \hat{u}_{\theta^*_{(1,2)}}(2,2) = 4\theta^*_{(1,2)}.$$

The total perceived difference is therefore $3\theta^*_{(1,2)}$. Since

$$\hat{u}_{\theta^*_{(1,2)}}(1,2) = \theta^*_{(1,2)},$$

the high type's IC constraint permits the markup

$$\hat{u}_{\theta^*_{(1,2)}}(2,2) - \hat{u}_{\theta^*_{(1,2)}}(1,2) = 3\theta^*_{(1,2)}.$$

The monopolist can therefore capture the entire perceived difference:

$$\frac{\hat{u}_{\theta^*_{(1,2)}}(2,2) - \hat{u}_{\theta^*_{(1,2)}}(1,2)}{\hat{u}_{\theta^*_{(1,2)}}(2,2) - \hat{u}_{\theta^*_{(1,2)}}(1,1)} = 1.$$

Thus, correct specification gives the monopolist no scope to extract the difference between types through quality screening, whereas the misspecified model permits the entire perceived difference to be captured through the IC markup.

Second, the dispersion in perceived utilities matters. Under the optimal pooling profile $(1,1)$, the fitting problem is

$$\min_{\theta \in \Theta} \left\{ \lambda(1-\theta)^2 + (1-\lambda)(2-\theta)^2 \right\}.$$

Both types purchase product 1, and the subjective model assigns the same utility θ to both purchases. The fitted parameter is therefore

$$\theta^*_{(1,1)} = 2 - \lambda \in (1,2).$$

Feedback from the high type raises the common fitted scale above the low type's true utility. Since this same scale determines the low type's perceived utility,

$$\hat{u}_{\theta^*_{(1,1)}}(1,1) = \theta^*_{(1,1)} > 1 = u^*(1,1),$$

the monopolist can charge

$$p_1 = \theta^*_{(1,1)} = 2 - \lambda$$

to every consumer.

The high type's greater experienced utility is therefore absorbed entirely through an increase in the common scale, which simultaneously raises the low type's perceived utility.

By contrast, under the separating profile $(1,2)$, the subjective model now imposes substantially greater dispersion than the true model. Consumers solve

$$\min_{\theta \in \Theta} \left\{ \lambda(1-\theta)^2 + (1-\lambda)(2-4\theta)^2 \right\},$$

giving

$$\theta^*_{(1,2)} = \frac{8 - 7\lambda}{16 - 15\lambda}.$$

As in Example 1, the sensitivity of 4θ causes the quadratic fit to prioritize the high type's experienced utility. The fitted scale is pulled towards $1/2$, causing the model to undervalue product 1 for the low type.

The dispersion force therefore favours pooling on product 1 over separation. Under pooling, high-type feedback raises the common perceived utility and spills over fully to low types. Under separation, the excessive dispersion imposed by the subjective model instead pushes down the fitted scale and reduces the perceived utility of the low-quality product.

This example illustrates the reverse of Example 1. Correct specification provides no profitable IC markup from quality screening, while exponent misspecification permits the full perceived difference to be captured. More importantly, pooling removes the perceived dispersion between types, allowing high-type feedback to raise the price paid by low types. The monopolist consequently earns strictly more and pools both types on the low-quality product for every non-degenerate type distribution.

4.4 Example 4: Two-Parameter Cobb–Douglas Misspecification

Throughout this example, let $M = \{0, 1, 2\}$, $V = \{1, 2\}$, and $\lambda := \lambda_1$. Set the true utility to

$$u^*(m, v) = \min\{m, v\}.$$

As in Example 1, low-type consumers obtain no additional utility from purchasing product 2, whereas high-type consumers obtain utility 1 from product 1 and utility 2 from product 2.

Correctly specified benchmark. Under correct specification, the monopolist separates the two types by setting

$$\mathbf{p}^* = (1, 2).$$

The resulting consumption profile is

$$\mathbf{m} = (1, 2),$$

and profit per consumer is

$$\Pi^* = 2 - \lambda.$$

Misspecified model. Consumers believe that utility belongs to the two-parameter family

$$\widehat{U} = \left\{ \hat{u}_{k,b}(m, v) : (k, b) \in \Theta \right\},$$

where

$$\hat{u}_{k,b}(m, v) = \begin{cases} 0, & m = 0, \\ k m^b v^{1-b}, & m > 0, \end{cases}$$

and

$$\Theta = [0, 2] \times [0, 1].$$

This is a regular subjective model with scale parameter k and shape parameter b. The parameter k determines the overall magnitude of perceived utility, whereas b determines the relative importance assigned to product quality and consumer type. When $b = 0$, consumers believe that utility depends only on type. When $b = 1$, they believe that utility depends only on product quality.

Weighted-KL minimisation under $\mathbf{m} = (1, 2)$. Consider the separating profile

$$\mathbf{m} = (1, 2).$$

Without experimentation, the wKLD fitting problem is

$$\min_{(k,b)\in\Theta} \left\{\lambda(1-k)^2 + (1-\lambda)(2-2k)^2\right\}.$$

The on-path observations uniquely identify

$$k^*_{(1,2)} = 1,$$

but the objective is independent of b. Hence, the set of unperturbed best-fitting parameters is

$$\{(1, b) : b \in [0, 1]\}.$$

The equilibrium observations therefore identify the scale of perceived utility but do not identify the relative importance of quality and type. The limiting value of b is instead selected by experimentation.

A low type who experiments with product 2 generates the observation

$$u^*(2, 1) = 1, \qquad \hat{u}_{1,b}(2, 1) = 2^b.$$

A high type who experiments with product 1 generates

$$u^*(1, 2) = 1, \qquad \hat{u}_{1,b}(1, 2) = 2^{1-b}.$$

Consequently, along the set of on-path minimisers, experimentation selects b by minimising

$$\lambda\eta^n_1(2^b - 1)^2 + (1-\lambda)\eta^n_2(2^{1-b} - 1)^2.$$

We now compare three vanishing experimentation paths.

Equal-rate experimentation. First, suppose

$$\eta^n = (\delta_n, \delta_n), \qquad \delta_n \downarrow 0.$$

After removing the common factor δ_n, the limiting fitting problem is

$$\min_{b\in[0,1]} \left\{\lambda(2^b - 1)^2 + (1-\lambda)(2^{1-b} - 1)^2\right\}.$$

Let b_λ denote its solution. The first-order condition is

$$\lambda 2^{b_\lambda}(2^{b_\lambda} - 1) = (1-\lambda)2^{1-b_\lambda}(2^{1-b_\lambda} - 1).$$

For every $\lambda \in (0,1)$, the solution is unique and satisfies

$$b_\lambda \in (0,1).$$

In particular, when $\lambda = 1/2$, symmetry gives

$$b_\lambda = \frac{1}{2}.$$

Under the limiting fit $(k,b) = (1, b_\lambda)$, the low type's perceived gain from purchasing product 2 instead of product 1 is

$$\hat{u}_{1,b_\lambda}(2,1) - \hat{u}_{1,b_\lambda}(1,1) = 2^{b_\lambda} - 1.$$

The corresponding perceived gain for the high type is

$$\hat{u}_{1,b_\lambda}(2,2) - \hat{u}_{1,b_\lambda}(1,2) = 2 - 2^{1-b_\lambda}.$$

For the low type to purchase product 1, individual rationality requires

$$\mathrm{IR}_L^{(1,2)}: \qquad p_1 \le \hat{u}_{1,b_\lambda}(1,1) = 1. \tag{18}$$

The low type must strictly prefer product 1 to product 2, whereas the high type may be indifferent because the tie-breaking rule selects product 2. Hence,

$$\mathrm{IC}_L^{(1,2)}: \qquad p_2 - p_1 > 2^{b_\lambda} - 1, \tag{19}$$

$$\mathrm{IC}_H^{(1,2)}: \qquad p_2 - p_1 \le 2 - 2^{1-b_\lambda}. \tag{20}$$

For every $b_\lambda \in (0,1)$,

$$2^{b_\lambda} + 2^{1-b_\lambda} < 3,$$

and therefore

$$2^{b_\lambda} - 1 < 2 - 2^{1-b_\lambda}.$$

Conditions (19) and (20) are therefore compatible, so the separating profile can be sustained.

The revenue-maximising supporting prices bind (18) and (20):

$$\mathbf{p}_{(1,2)} = \left(1,\, 3 - 2^{1-b_\lambda}\right).$$

Conditional on the separating profile being selected, the resulting profit is

$$\pi_{(1,2)}^{\text{equal}} = \lambda + (1-\lambda)\left(3 - 2^{1-b_\lambda}\right).$$

This is strictly below the correctly specified benchmark because

$$\Pi^* - \pi_{(1,2)}^{\text{equal}} = (1-\lambda)\left(2^{1-b_\lambda} - 1\right) > 0.$$

Thus, equal-rate experimentation permits screening, but the monopolist charges the high type less than under correct specification.

Low-type experimentation vanishes faster. Next, suppose

$$\eta^n = (\delta_n^2, \delta_n).$$

Experimental observations generated by high types receive asymptotically greater weight than those generated by low types. The limiting fitting problem therefore selects

$$b^* = 1,$$

because this makes

$$\hat{u}_{1,1}(1,2) = 1 = u^*(1,2).$$

At this fit, both types perceive the same gain from product 2:

$$\hat{u}_{1,1}(2,1) - \hat{u}_{1,1}(1,1) = 1$$

and

$$\hat{u}_{1,1}(2,2) - \hat{u}_{1,1}(1,2) = 1.$$

The incentive constraints would therefore require

$$p_2 - p_1 > 1$$

for the low type to choose product 1, but

$$p_2 - p_1 \leq 1$$

for the high type to choose product 2. These conditions are incompatible, so $(1,2)$ cannot be sustained as a pBNE relative to this experimentation path.

High-type experimentation vanishes faster. Finally, suppose

$$\eta^n = (\delta_n, \delta_n^2).$$

Low-type experimental observations now receive asymptotically greater weight, and the limiting fitting problem selects

$$b^* = 0.$$

At this fit, consumers believe that utility depends only on type. Both types therefore perceive no gain from purchasing product 2 instead of product 1:

$$\hat{u}_{1,0}(2,v) - \hat{u}_{1,0}(1,v) = 0 \qquad \text{for every } v \in V.$$

The low-type and high-type incentive constraints become

$$p_2 - p_1 > 0 \qquad \text{and} \qquad p_2 - p_1 \leq 0,$$

respectively. They are again incompatible, so $(1,2)$ cannot be sustained as a pBNE relative to this experimentation path.

Result 4 (Experimentation-dependent screening)**.** *Under two-parameter Cobb–Douglas misspecification, the sustainability of the separating profile* $(1,2)$ *depends on the vanishing experimentation path.*

(i) Under equal-rate experimentation, the limiting fit satisfies $b_\lambda \in (0,1)$*, and the profile* $(1,2)$ *can be sustained. Its revenue-maximising supporting prices are*

$$\left(1,\, 3 - 2^{1-b_\lambda}\right),$$

and its conditional profit is strictly below the correctly specified benchmark.

(ii) If either type's experimentation rate vanishes one order faster than the other's, the limiting fit selects $b = 0$ *or* $b = 1$*, and the profile* $(1,2)$ *cannot be sustained.*

Economic intuition. The two-parameter model separates the fitting of scale from the fitting of shape. Under $(1,2)$, ordinary market feedback identifies $k = 1$, but it provides no information about b. The rare experimental observations therefore determine how consumers perceive the relative importance of product quality and consumer type.

When experimentation by the two types vanishes at the same rate, neither type's experimental feedback dominates. The fitted value of b lies in the interior of the parameter space, causing the high type to perceive a strictly larger gain from quality than the low type. The monopolist can then separate the two types, although the high type pays less than under correct specification.

When one type's experimentation vanishes faster, the observations generated by the other type dominate the selection of b. The fitted parameter moves to a boundary, and the two types perceive the same gain from quality. The strict low-type incentive constraint and the weak high-type incentive constraint then become incompatible. Thus, with more than one subjective parameter, the relative rates of experimentation can determine both the price used for screening and whether screening is possible at all.

4.5 Constant Models as Always-Winners

From the previous examples, higher dispersion in the misspecified model tends to push down the fitted values under a quadratic fit, this sheds light on a fact that whether the monopolist can beat the rational benchmark ultimately depends on how flat the relevant utility mapping is.

This subsection characterises when misspecification *uniformly* benefits the monopolist. Call a misspecified environment $\widehat{U}$ an *always-winner* if the monopolist's robust profit under $\widehat{U}$ is weakly higher than the correctly specified benchmark for every admissible u^*.

Next, I will state a corollary and prove it. This shows that when u^* is constant on the relevant cells, the monopolist cannot do better than the correctly specified benchmark. Hence a constant u^* serves as a worst-case test for any candidate always-winner misspecified family.

Corollary 1. *Fix the primitives* $\left\{M, V, \lambda, u^*, \widehat{U}, \varepsilon_1\right\}$*. Suppose the true utility* u^* *satisfies:*

$$u^*(1,1) = u^*(2,1) = u^*(1,2) = u^*(2,2)$$

As long as $\widehat{U}$ *is scale-invariant, monotonically increasing and single-crossing, monopolist can never set a price to beat the profit in the rational benchmark.*

Proof. WLOG assume the true utility returns constant value 1. Then the rational benchmark profit for monopolist is 1. We will show that the profit in misspecified environment is weakly lower than 1.

We first prove the single-template case. We can then represent the misspecified utility in an equilibrium as $k\hat{u} \in \widehat{U}$, where k is some constant. WLOG assumes:

$$\hat{u}(1,1) = a, \quad \hat{u}(2,1) = b, \quad \hat{u}(1,2) = c, \quad \hat{u}(2,2) = d$$

where a, b, c, d are chosen such that $\hat{u}$ is monotonically increasing and single-crossing. We could do this because k is the only free parameter in the misspecified environment.

Now let's check every equilibrium one by one. For the equilibrium $\mathbf{m} = (2,2)$, by checking the wKLD:

$$\lambda(kd-1)^2 + (1-\lambda)(kb-1)^2$$

Since $b \leq d$, the FOC must have $kb \leq 1$ or else kb and kd are both larger than 1, which means k can be lowered to further minimise the objective. While the profit is kb in this scenario, we know that $\mathbf{m} = (2,2)$ cannot achieve better than the benchmark.

For the equilibrium $\mathbf{m} = (1, 1)$, the same applies with b and d replaced by a and c.

For the equilibrium $\mathbf{m} = (0, 2)$, the consumers can pin down the true utility so the profit is no better than the benchmark.

For the equilbrium $\mathbf{m} = (1, 2)$, the wKLD:

$$\lambda(kd - 1)^2 + (1 - \lambda)(ka - 1)^2$$

The FOC is:

$$k^* = \frac{\lambda d + (1 - \lambda)a}{\lambda d^2 + (1 - \lambda)a^2}$$

The profit from $\mathbf{m} = (1, 2)$ is

$$k^*[a + \lambda(d - c)]$$

The upper bound of this profit is when c is the smallest. By assumption it is $c = a$, so we have the upper bound of the profit written as:

$$k^*[(1 - \lambda)a + \lambda d] = \frac{[\lambda d + (1 - \lambda)a]^2}{\lambda d^2 + (1 - \lambda)a^2}$$

The numerator of the R.H.S. is simply the square of a weighted mean, and its denominator is the weighted mean of the square. By Jensen's inequality, the fraction is never larger than 1.

Now check the case where $\widehat{U}$ is not single-template.

Consider an arbitrary misspecified family $\widehat{U}$ that is scale-invariant, monotone, and single-crossing. Fix a candidate profile m. The wKLD criterion selects (possibly set-valued) minimisers that best fit the *observation-cell* utilities. There are two cases.

Case 1: the observation cells pin down the truth. Suppose there exists a minimiser $\hat{u}$ such that the observation cells are fitted exactly, i.e. $\hat{u}(m_1, 1) = u^*(m_1, 1) = 1$ and $\hat{u}(m_2, 2) = u^*(m_2, 2) = 1$. Then IR immediately implies $p_1 \leq 1$ and $p_2 \leq 1$ along the equilibrium path. Moreover, in the separating profile $m = (1, 2)$, perfect fit yields $\hat{u}(1, 1) = \hat{u}(2, 2) = 1$. Under monotonicity, separation cannot generate a positive markup. Hence no equilibrium under perfect fit can yield profit above the correctly specified benchmark.

Case 2: the truth is not pinned down at the observation cells. By the regular utility assumption, the misspecified family is scale-invariant: if a utility function $\hat{u}$ is feasible, then so is $c\,\hat{u}$ for any $c > 0$. Hence we can parameterise $\hat{u}$ as

$$\hat{u}(m, v) = k\,\tilde{u}_\theta(m, v),$$

where $k > 0$ is a free scale parameter and θ collects all remaining (non-scale) parameters in the misspecified model. Fix a profile $m = (m_1, m_2)$ and define the two observation-cell predictions under $\tilde{u}_\theta$ as

$$x(\theta) := \tilde{u}_\theta(m_1, 1), \qquad y(\theta) := \tilde{u}_\theta(m_2, 2).$$

Since $u^*(m_1, 1) = u^*(m_2, 2) = 1$, the wKLD objective on the observation cells is

$$L(k; \theta) = \lambda\big(1 - kx(\theta)\big)^2 + (1 - \lambda)\big(1 - ky(\theta)\big)^2,$$

which is a strictly convex quadratic in k. Therefore, conditional on θ, the unique best-fit scale is

$$k^*(\theta) = \frac{\lambda x(\theta) + (1 - \lambda)y(\theta)}{\lambda x(\theta)^2 + (1 - \lambda)y(\theta)^2}.$$

If $x(\theta) = y(\theta)$, the fit is exact. Otherwise, fitting the constant target $(1, 1)$ with the scaled pair $(kx(\theta), ky(\theta))$ necessarily creates dispersion across the observation cells, and the quadratic criterion penalises this dispersion. This is the same mechanism as in the single-template case: we can still use the Jensen's argument, so the monopolist cannot guarantee profit above the correctly specified benchmark.

□

Conversely, this next corollary shows that if $\hat{u}$ is constant (up to scale) across the four relevant cells, then the monopolist weakly beats the benchmark for any u^*, with strict inequality generically.

Corollary 2. *Fix the primitives $\{M, V, \lambda, u^*, \hat{U}, \hat{\varepsilon}_1\}$. Suppose the misspecified utility function $\hat{u}$ is scale-invariant and maps (m, v) such that*

$$\hat{u}(1, 1) = \hat{u}(2, 1) = \hat{u}(1, 2) = \hat{u}(2, 2).$$

Then the monopolist can set a price to weakly *beat the rational benchmark, and the inequality is strict unless $u^*(2, 1) = u^*(1, 2) = u^*(1, 1)$ (regardless of $u^*(2, 2)$). Otherwise, the monopolist can match the rational-benchmark profit.*

Proof. Note that with this misspecified model, the profit is going to be exactly equal to the scaling constant k in the equilibrium for the misspecified model.

First, write the true utility mapping as:

$$u^*(1, 1) = a, \quad u^*(2, 1) = b, \quad u^*(1, 2) = c, \quad u^*(2, 2) = d.$$

In the correctly specified benchmark, the monopolist earns

$$\max\{\, b,\; a + \lambda(d - c),\; \lambda d \,\}.$$

Now let's check every equilibrium one by one. Again, we omit checking the equilibrium $(0, 2)$: since consumers can pin down the true utility there, the profit is no better than the benchmark.

Start with $m = (2, 2)$. The wKLD is

$$\lambda(k - d)^2 + (1 - \lambda)(k - b)^2.$$

The FOC gives the unique minimiser

$$k^* = \lambda d + (1 - \lambda)b,$$

and the profit from the equilibrium is therefore k^*.

For $m = (1, 1)$, the wKLD is

$$\lambda(k - c)^2 + (1 - \lambda)(k - a)^2,$$

so the FOC gives $k^* = \lambda c + (1 - \lambda)a$.

Finally, for $m = (1, 2)$, the wKLD is

$$\lambda(k - d)^2 + (1 - \lambda)(k - a)^2,$$

so the FOC gives $k^* = \lambda d + (1 - \lambda)a$.

Note that the profit from $(2, 2)$ is weakly the highest, since $b \geq a$ by monotonicity. Thus it suffices to compare $(2, 2)$ with the benchmark:

$$\lambda d + (1 - \lambda)b \;\geq\; \max\{\, b,\; a + \lambda(d - c),\; \lambda d \,\}.$$

The left-hand side is a convex combination of d and b, so it dominates b and λd. It remains to show

$$\lambda d + (1 - \lambda)b \geq a + \lambda(d - c) \iff (1 - \lambda)b \geq a - \lambda c \iff \lambda c + (1 - \lambda)b \geq a.$$

By monotonicity, $b = u^*(2, 1) \geq u^*(1, 1) = a$ and $c = u^*(1, 2) \geq a$, so the left-hand side is a convex combination of two numbers weakly larger than a. Therefore $\lambda c + (1 - \lambda)b \geq a$, and the misspecified profit weakly exceeds the rational benchmark. Equality obtains iff $b = c = a$ (regardless of d). $\square$

Putting these two corollaries together yields a sharp dominance result: the constant misspecified model is the unique always-winner in our environment.

Proposition 1 (The constant model is the unique robust winner)**.** *Fix primitives* $\{M, V, \lambda, \hat{\varepsilon}_1\}$ *and maintain the utility assumptions on* U *(in particular, scale-indispersion, monotonicity, and single-crossing).*

*(i) (*Sufficiency*) If for all $\hat{u} \in \widehat{U}$,*

$$\hat{u}(1,1) = \hat{u}(2,1) = \hat{u}(1,2) = \hat{u}(2,2),$$

then $\widehat{U}$ is an always-winner. Moreover, the robust profit is strictly higher than the correctly specified benchmark unless $u^*(2,1) = u^*(1,2) = u^*(1,1)$.

*(ii) (*Unique Always-Winner*) Conversely, suppose $\widehat{U}$ does* not *contain any misspecified utility that is constant on these four cells. Then $\widehat{U}$ is* not *an always-winner: there exists an admissible true utility u^* (in particular, a constant u^* on the four relevant cells) such that the correctly specified benchmark profit strictly exceeds the monopolist's robust profit under $\widehat{U}$.*

Proof. Part (i) is Corollary 2. For part (ii), take u^* constant on the four relevant cells and apply Corollary 1: any non-constant misspecified family fails to weakly dominate the correctly specified benchmark in this environment. □

5 Conclusion and Future Directions

This paper studies monopoly pricing when consumers learn from market feedback using a misspecified utility model. Because consumers observe satisfaction only from products that are purchased, prices affect not only demand but also the data used to estimate utility. The monopolist consequently faces an interaction between pricing, consumer learning, and equilibrium selection.

The examples show that misspecification does not have a uniform effect on profit. It may reduce profit by lowering consumers' fitted utilities or by restricting the incentive-compatible markup available under separation. However, it may also benefit the monopolist when feedback from consumers with high realised utility raises the fitted value of products purchased by other consumers. In particular, misspecification can make pooling on the low-quality product profitable even though this allocation is never optimal under correct specification.

Two forces are central to these comparisons. The first is the incentive-compatible markup implied by perceived utilities. Even when misspecification raises consumers' willingness to pay, the monopolist can extract the increase through screening only if the high type perceives a sufficiently larger gain from quality than the low type. The second is the dispersion of predicted utilities across the observation cells used in the wKLD fit. Under a quadratic fitting criterion, a subjective model that imposes greater dispersion than the true utility tends to fit some observations downward, whereas a flatter model may allow high realised utility to raise the common fitted scale.

Robust equilibrium selection introduces an additional consideration. A price vector designed to support a profitable profile may also support another pBNE profile associated with a different set of observations and a different fitted belief. The monopolist must therefore account for these coexisting profiles rather than evaluating prices only under the intended allocation. This is why profiles that cannot themselves generate optimal profit may nevertheless affect the monopolist's robust objective.

The two-parameter example shows that the experimentation path can also matter. With more than one subjective parameter, equilibrium observations may identify the overall scale of utility without identifying how utility responds to product quality and consumer type. Rare experimental observations then select the remaining parameter. As a result, the relative rates at which different types experiment may determine the supporting prices and even whether a separating profile exists. This role of experimentation is absent when the on-path observations already identify the subjective parameter uniquely.

General conditions for profitable misspecification. A natural next step is to replace the profile-specific calculations with general sufficient conditions under which misspecification benefits the monopolist. For a regular single-template model

$$\hat{u}_k(m, v) = kg(m, v),$$

the examples suggest that such conditions must combine three objects: the dispersion of g across the relevant observation cells, the incentive-compatible markup generated by g, and the prices required to eliminate less profitable coexisting profiles.

For the separating profile $(1, 2)$, lower dispersion at the observation cells may raise the fitted scale, but this is profitable only if the resulting perceived quality increment is sufficiently larger for the high type than for the low type. The supporting price must also exclude profiles such as $(0, 1)$, which may otherwise reduce robust profit. For the pooling profile $(2, 2)$, the relevant question is whether the fitted utility of the low type exceeds its correctly specified value while the supporting prices exclude profiles such as $(0, 2)$. Formalising these requirements would provide a general sufficient condition for misspecification to raise robust profit.

Experimentation with multidimensional models. The analysis also leaves open how equilibrium should be selected when the subjective model contains several parameters and different experimentation paths produce different limiting beliefs. One possibility is to require the monopolist's prices to be robust across a class of vanishing experimentation paths. Another is to endogenise experimentation by allowing its frequency to depend on consumer type, prior experience, or the perceived cost of trying an unfamiliar product. Either extension would clarify when the path dependence identified in the two-parameter example has economically significant effects on monopoly pricing.

Larger menus and mixed consumption. The present equilibrium definition restricts attention to pure consumption profiles. In particular, the $\max\arg\max$ tie-breaking rule assigns every consumer of a given type to the highest product among their optimal choices. Thus, if type v is indifferent between products 2 and 3, all consumers of that type choose product 3. This is reflected in the strict incentive constraint required for a type to select the lower product. Consequently, mixed consumption is ruled out under the current definition.

A possible extension would enlarge the menu, for example to

$$M = \{0, 1, 2, 3\},$$

and allow consumers of the same type to randomise across optimal products. Low types might purchase product 1, while some high types purchase product 2 and the remaining high types purchase product 3. Such an allocation would require

$$\bar{u}_\mu(2,2) - p_2 = \bar{u}_\mu(3,2) - p_3,$$

with both products belonging to the high type's set of optimal choices.

This extension may be economically relevant under misspecification. If high types purchase both products 2 and 3, the observation cells

$$(2,2) \qquad \text{and} \qquad (3,2)$$

both receive non-vanishing weight in the wKLD objective. The resulting belief may therefore differ from the belief generated when all high types purchase only one product. Although mixed consumption is generally unnecessary in the correctly specified two-type Mussa–Rosen benchmark, it may affect profit here because it changes the market feedback used to fit the subjective model.

Studying this possibility would require extending the definition of pBNE to include type-specific mixed strategies and making the fitted belief depend on the resulting endogenous choice frequencies.

References

Clyde, A. (2025). Narrow inference and incentive design.

Eliaz, K., & Spiegler, R. (2006). Contracting with diversely naive agents. *The Review of Economic Studies*, *73*(3), 689–714. doi: doi:10.1111/j.1467-937X.2006.00392.x

Esponda, I., & Pouzo, D. (2016). Berk–nash equilibrium: A framework for modeling agents with misspecified models. *Econometrica*, *84*(3), 1093–1130. doi: doi:10.3982/ECTA12609

Heidhues, P., & Kőszegi, B. (2018). Behavioral industrial organization. In B. D. Bernheim, S. DellaVigna, & D. Laibson (Eds.), *Handbook of behavioral economics: Applications and foundations 1* (Vol. 1, pp. 517–612). Elsevier. doi: doi:10.1016/bs.hesbe.2018.07.006

Heidhues, P., Kőszegi, B., & Murooka, T. (2017). Inferior products and profitable deception. *The Review of Economic Studies*, *84*(1), 323–356. doi: doi:10.1093/restud/rdw037

Ispano, A., & Schwardmann, P. (2023). Cursed consumers and the effectiveness of consumer protection policies. *The Journal of Industrial Economics*, *71*(2), 407–440. doi: doi:10.1111/joie.12332

Kőszegi, B. (2014). Behavioral contract theory. *Journal of Economic Literature*, *52*(4), 1075–1118. doi: doi:10.1257/jel.52.4.1075

Murooka, T., & Yamashita, T. (2023). Adverse selection and bounded rationality: An impossibility theorem. *The Japanese Economic Review*, *74*(3), 439–444. doi: doi:10.1007/s42973-022-00119-w

Mussa, M., & Rosen, S. (1978). Monopoly and product quality. *Journal of Economic Theory*, *18*(2), 301–317. doi: doi:10.1016/0022-0531(78)90085-6

Spiegler, R. (2011). *Bounded rationality and industrial organization.* Oxford University Press.